Supplementary Material

Toward Robust Post-Selection Inference in Linear Mixed Models:

An Evaluation and Extension of the *selfmade* Approach

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Supplementary Material

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Overview

This supplementary material follows the structure of the main paper and provides additional technical details, formal proofs, and complementary simulation results. We first present pseudocode for our implementation of the selfmade approach. We then establish the equivalence relations underlying the computation of p-values within the selfmade approach. We then provide details on the use of importance sampling to improve the computational efficiency of the MC sampling procedure in the selfmade approach. Next, we prove the proposition discussed in Simulation Study 1, which helps explain the performance of the frequentist conditional perspective. We then derive the inference targets for the working models considered in Simulation Studies 2–4 under the efficient marginal, not-efficient marginal, and conditional perspectives, and explain why the simulation design does not align with the assumptions underlying the conditional perspective. Next, we report the convergence issues encountered in Simulation Studies 1–4 in greater detail. We then provide proofs

concerning the singularity properties of the new covariance matrix estimators proposed in Simulation Study 3, using the marginal perspective as an example. We further present a repetition of Simulation Study 3 under the uncorrected Bayesian perspective, and a proof that all conditional perspectives essentially coincide for the fixed effects considered in Simulation Study 3. Finally, we describe an additional simulation study that closely resembles Simulation Study 3 but generates the predictor matrix randomly rather than treating it as fixed.

Pseudocode: Selfmade

This section presents pseudocode for our implementation of the selfmade approach, including the estimation of the required covariance matrices and the computation of the parameters of the test statistic.

The algorithm is divided into four parts. Part (a) performs model selection and extracts the relevant design information. Part (b) estimates the required covariance matrices based on the chosen inferential perspective and estimation strategy. Part (c) uses these estimates to compute the quantities that define the test statistic. Finally, Part (d) implements the MC sampling procedure used to obtain post-selection p-values and confidence intervals.

The complete implementation is available in the project's GitHub repository.

Algorithm 1: Selfmade - Input and Output

Input:

Response vector $\mathbf{y}_{\text{obs}}$ and candidate predictor matrix $\mathbf{X}$

Selection procedure $\mathcal{S}$

Assumptions on the random-effects structure and on $\mathbf{R}$

Inferential perspective (frequentist conditional, uncorrected Bayesian conditional, corrected Bayesian conditional, efficient marginal or not-efficient marginal)

Covariance matrix estimator (*ModelEstimate, ICM, VarY*, if applicable)

Whether $\boldsymbol{Q}$ replaces $\mathbf{R}$ or $\boldsymbol{\Sigma}$, if applicable

Choice of HC correction and cluster adjustment, if applicable

If and which CR-based variance estimation should be used, if applicable

Name $name_{p_j}$ of the selected model term p_j for which selective inference is required

Value of ρ under H_0

Number of MC samples B

MC distribution $\mathcal{D}$ ($\mathcal{D} = \mathcal{N}(0, \kappa^2)$ or $\mathcal{D} = \mathcal{N}(t_{\text{obs}}, \kappa^2)$)

Should convergence of fitted models be checked in MC Sampling step?

Significance level α for inference

Output:

Post-selection p-value

$(1 - \alpha)$ Confidence interval for effect of selected predictor

Algorithm 2: Selfmade - Meta-Algorithm

Part (a): Conduct model selection and extract the selected model using Algorithm 3

if *any marginal perspective* **then**

  if $\boldsymbol{\Sigma}$ *is known* **then**

    Use supplied covariance matrix

  else

    Part (b): Obtain covariance matrix estimate for $\boldsymbol{\Sigma}$ using Algorithm 4

if *any conditional perspective* **then**

  if $\boldsymbol{R}, \boldsymbol{G}_{working}$ *are known* **then**

    Use supplied covariance matrices

  else

    Part (b): Obtain covariance matrix estimates for $\boldsymbol{R}, \boldsymbol{G}_{\text{working}}$ using Algorithm 4

Part (c): Use Algorithm 7. Compute quantities of the test statistic (test vector, variance of the test statistic, direction vector, and the component of the outcome vector orthogonal to the test vector) using the covariance matrices (or their estimates).[a]

Part (d): Sample from sampling distribution conditional on the selection event and compute p-values and confidence intervals using Algorithm 9.

[a] For simplicity, we use $\boldsymbol{\Sigma}$, $\boldsymbol{R}$, and $\boldsymbol{G}_{\text{working}}$ also for their estimates.

Algorithm 3: Selfmade - Part (a): Model Selection and Design Extraction

Select model $\mathcal{M}$ using $\mathcal{S}$, extract number of selected predictors P and obtain initial selection outcome $(\mathcal{A}, \mathcal{B}) = \mathcal{S}(\mathbf{y}_{\text{obs}})$

Extract selected fixed-effects design matrix $\mathbf{X}_{\mathcal{A}}$ from $\mathcal{M}$

Extract random-effects design matrix $\mathbf{Z}_{\mathcal{B}}$ from $\mathcal{M}$

Identify the column index j of $name_{p_j}$ from $(\mathbf{X}_{\mathcal{A}}, \mathbf{Z}_{\mathcal{B}})$

Extract number of observations N, number of Level-2 units M and Level-2 unit identity from $\mathcal{M}$

Algorithm 4: Selfmade - Part (b): Estimation of Covariance Matrices

Extract the random effects covariance matrix $\hat{\mathbf{G}}_{\mathcal{M}}$ from $\mathcal{M}$

if *(random-effects structure is known and* $\mathbf{R} = \sigma^2_{\varepsilon}\mathbf{I}_N$ *is assumed) or (CR based variance estimation is used)* **then**

  Proceed with Algorithm 5

else

  Use the not-efficient marginal perspective and proceed with Algorithm 6

Algorithm 5: Selfmade - Part (b.1)

Estimate $\hat{\mathbf{R}}$ using one of the following:

ModelEstimate: Extract the Level-1 residual covariance matrix $\hat{\mathbf{R}} = \hat{\mathbf{R}}_{\mathcal{M}}$ from the selected model $\mathcal{M}$

ICM: Fit the Intercept Model $\mathbf{y}_{\text{obs}} = \hat{\boldsymbol{\beta}}_0^{ICM}\mathbf{1}_N + \mathbf{Z}_{\mathcal{B}}\hat{\mathbf{b}}^{ICM} + \hat{\boldsymbol{\varepsilon}}^{ICM}$ and set $\hat{\mathbf{R}} = \sigma^2_{\hat{\varepsilon}^{ICM}}\mathbf{I}_N$

VarY: $\hat{\mathbf{R}} = \text{var}(\mathbf{y}_{\text{obs}}) \cdot \frac{N-1}{N}\mathbf{I}_N$

if *conditional perspective* **then**

  $\hat{\boldsymbol{G}}_{\text{working}} = \hat{\mathbf{G}}_{\mathcal{M}}$

if *marginal perspective* **then**

  if *ModelEstimate* **then**

    $\hat{\boldsymbol{\Sigma}} = \mathbf{Z}_{\mathcal{B}}\hat{\mathbf{G}}_{\mathcal{M}}\mathbf{Z}_{\mathcal{B}}^{\top} + \hat{\mathbf{R}}$

  else if *ICM* **then**

    $\hat{\boldsymbol{\Sigma}} = \mathbf{Z}_{\mathcal{B}}\hat{\mathbf{G}}^{ICM}\mathbf{Z}_{\mathcal{B}}^{\top} + \hat{\mathbf{R}}$, where $\hat{\mathbf{G}}^{ICM}$ is the random-effects covariance matrix estimated from the fitted Intercept Model

Algorithm 6: Selfmade - Part (b.2)

Compute non-spherical covariance matrix estimate $\boldsymbol{Q}$ to replace $\boldsymbol{\Sigma}$ or $\mathbf{R}$:

if *ModelEstimate* **then**

Extract residuals $\hat{\boldsymbol{\varepsilon}}^{LM}$ from linear model $\mathbf{y}_{\text{obs}} = \mathbf{X}_{\mathcal{A}}\hat{\boldsymbol{\beta}}^{LM} + \hat{\boldsymbol{\varepsilon}}^{LM}$

Compute $\mathbf{H} = \mathbf{X}_{\mathcal{A}}(\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A}}^{\top}$

else if *ICM* **then**

Extract residuals $\hat{\boldsymbol{\varepsilon}}^{LM}$ from linear intercept model $\mathbf{y}_{\text{obs}} = \bar{\mathbf{y}}_{\text{obs}}\mathbf{1}_N + \hat{\boldsymbol{\varepsilon}}^{LM}$

Compute $\mathbf{H} = \frac{\mathbf{1}_N\mathbf{1}_N^{\top}}{N}$

Compute HC covariance matrix (Level-2 units $m = 1, \ldots, M$):

$\boldsymbol{Q}_{HC} = \boldsymbol{Q}_{HC0} = \text{diag}(\hat{\boldsymbol{\varepsilon}}_1^{LM}(\hat{\boldsymbol{\varepsilon}}_1^{LM})^{\top}, \ldots, \hat{\boldsymbol{\varepsilon}}_M^{LM}(\hat{\boldsymbol{\varepsilon}}_M^{LM})^{\top})$

if *HC1* **then**

$\boldsymbol{Q}_{HC} = \frac{N-1}{N-P-1}\boldsymbol{Q}_{HC0}$

else if *HC2* **then**

$\boldsymbol{Q}_{HC} = \boldsymbol{Q}_{HC0} \oslash (\mathbf{1} - \boldsymbol{H})$, where $\oslash$ denotes element-wise division

else if *HC3* **then**

$\boldsymbol{Q}_{HC} = \frac{M-1}{M} \cdot \boldsymbol{Q}_{HC0} \oslash ((\mathbf{1} - \boldsymbol{H}) \circ (\mathbf{1} - \boldsymbol{H}))$, where $\circ$ denotes element-wise multiplication

if *cluster adjustment required* **then**

$\boldsymbol{Q} = \frac{M}{M-1}\boldsymbol{Q}_{HC}$

else

$\boldsymbol{Q} = \boldsymbol{Q}_{HC}$

if $\boldsymbol{\Sigma}$ *should be replaced by* $\boldsymbol{Q}$ **then**

$\hat{\boldsymbol{\Sigma}} = \boldsymbol{Q}$

else if $\mathbf{R}$ *should be replaced by* $\boldsymbol{Q}$ **then**

$\hat{\boldsymbol{\Sigma}} = \mathbf{Z}_{\mathcal{B}}\hat{\mathbf{G}}_{\mathcal{M}}\mathbf{Z}_{\mathcal{B}}^{\top} + \boldsymbol{Q}$

Algorithm 7: Selfmade - Part (c): Test Statistic Parameters

if *conditional perspective* **then**

Define $\mathbf{C} = (\mathbf{X}_{\mathcal{A}}, \mathbf{Z}_{\mathcal{B}})$

Define $\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 0 & \boldsymbol{G}^{-1}_{\text{working}} \end{pmatrix}$

Calculate test vector $\mathbf{v} = \mathbf{R}^{-1}\mathbf{C}(\mathbf{C}^\top\mathbf{R}^{-1}\mathbf{C} + \mathbf{A})^{-1}\mathbf{e}_j$

Calculate variance of test statistic:

if *Bayesian* **then**

$\kappa^2 = \mathbf{e}_j^\top(\mathbf{C}^\top\mathbf{R}^{-1}\mathbf{C} + \mathbf{A})^{-1}\mathbf{e}_j$

else if *frequentist* **then**

$\kappa^2 = \mathbf{v}^\top\mathbf{R}\mathbf{v}$

Calculate $\mathbf{dir}_\mathbf{v}$:

if *corrected Bayesian* **then**

$\mathbf{dir}_\mathbf{v} = \frac{\mathbf{R}\mathbf{v}}{\mathbf{v}^\top\mathbf{R}\mathbf{v}}$

else

$\mathbf{dir}_\mathbf{v} = \frac{\mathbf{R}\mathbf{v}}{\kappa^2}$

else

if *efficient marginal perspective* **then**

Calculate test vector $\mathbf{v} = \boldsymbol{\Sigma}^{-1}\mathbf{X}_{\mathcal{A}}(\mathbf{X}_{\mathcal{A}}^\top\boldsymbol{\Sigma}^{-1}\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{e}_j$

else if *not-efficient marginal perspective* **then**

Calculate test vector $\mathbf{v} = \mathbf{X}_{\mathcal{A}}(\mathbf{X}_{\mathcal{A}}^\top\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{e}_j$

Calculate $\mathbf{dir}_\mathbf{v} = \frac{\boldsymbol{\Sigma}\mathbf{v}}{\mathbf{v}^\top\boldsymbol{\Sigma}\mathbf{v}}$

Calculate variance of test statistic $\kappa^2 = \mathbf{v}^\top\boldsymbol{\Sigma}\mathbf{v}$

if *CR-based variance estimation* **then**

Compute $\widehat{\mathbf{K}}_{CR}$ using Algorithm 8 and replace κ^2 by $\mathbf{e}_j^\top\widehat{\mathbf{K}}_{CR}\,\mathbf{e}_j$

Compute observed test statistic $t_{\text{obs}} = \mathbf{v}^\top\mathbf{y}_{\text{obs}}$

Compute $P_v^\perp(\mathbf{y}_{\text{obs}}) = (\mathbf{I}_N - \mathbf{dir}_\mathbf{v} \cdot \mathbf{v}^\top)\mathbf{y}_{\text{obs}}$

Algorithm 8: Selfmade - Part (c.2): CR-Based Variance Estimation

if *not-efficient marginal perspective* **then**

- $\mathbf{Bread} = (\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}$
- $\hat{\boldsymbol{\beta}}_{\mathcal{A}} = \mathbf{Bread}\,\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{y}_{\text{obs}}$
- $\hat{\boldsymbol{\varepsilon}}^{\text{LM}} = \mathbf{y}_{\text{obs}} - \mathbf{X}_{\mathcal{A}}\hat{\boldsymbol{\beta}}_{\mathcal{A}}$
- **for** *each cluster* $m = 1, \ldots, M$ **do**
  - Extract $\mathbf{X}_{\mathcal{A},m}$ and $\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}}$
  - **if** *CR0* **then**
    - $\mathbf{u}_m = (\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}})^{\top}\mathbf{X}_{\mathcal{A},m}$
  - **else if** *CR2* **then**
    - $\mathbf{H}_m = \mathbf{X}_{\mathcal{A},m}\,\mathbf{Bread}\,\mathbf{X}_{\mathcal{A},m}^{\top}$
    - $\mathbf{Adj}_m = (\mathbf{I}_{N_m} - \mathbf{H}_m)^{-1/2}$
    - $\mathbf{u}_m = (\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}})^{\top}\mathbf{Adj}_m\mathbf{X}_{\mathcal{A},m}$

if *efficient marginal perspective* **then**

- $\mathbf{Bread} = (\mathbf{X}_{\mathcal{A}}^{\top}\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{X}_{\mathcal{A}})^{-1}$, $\hat{\boldsymbol{\beta}}_{\mathcal{A}} = \mathbf{Bread} \cdot \mathbf{X}_{\mathcal{A}}^{\top}\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{y}_{\text{obs}}$, $\hat{\boldsymbol{\varepsilon}}^{\text{LM}} = \mathbf{y}_{\text{obs}} - \mathbf{X}_{\mathcal{A}}\hat{\boldsymbol{\beta}}_{\mathcal{A}}$
- **for** *each cluster* $m = 1, \ldots, M$ **do**
  - Extract $\mathbf{X}_{\mathcal{A},m}$, $\hat{\boldsymbol{\Sigma}}_m$, and $\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}}$
  - **if** *CR0* **then**
    - $\mathbf{u}_m = (\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}})^{\top}\hat{\boldsymbol{\Sigma}}_m^{-1}\mathbf{X}_{\mathcal{A},m}$
  - **else if** *CR2* **then**
    - $\mathbf{H}_m = \mathbf{X}_{\mathcal{A},m} \cdot \mathbf{Bread} \cdot \mathbf{X}_{\mathcal{A},m}^{\top}\hat{\boldsymbol{\Sigma}}_m^{-1}$
    - Compute Cholesky factor $\mathbf{U}_m$ such that $\hat{\boldsymbol{\Sigma}}_m = \mathbf{U}_m^{\top}\mathbf{U}_m$
    - $\mathbf{B}_m = \mathbf{U}_m\left[\mathbf{I}_{N_m} - \mathbf{H}_m\right]\hat{\boldsymbol{\Sigma}}_m\,\mathbf{U}_m^{\top}$
    - $\mathbf{Adj}_m = \mathbf{U}_m^{\top}\,\mathbf{B}_m^{-\frac{1}{2}}\,\mathbf{U}_m$
    - $\mathbf{u}_m = (\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}})^{\top}\mathbf{Adj}_m\,\hat{\boldsymbol{\Sigma}}_m^{-1}\mathbf{X}_{\mathcal{A},m}$

$\widehat{\mathbf{K}}_{CR} = \mathbf{Bread} \cdot \mathbf{Meat} \cdot \mathbf{Bread}$, with $\mathbf{Meat} = \sum_{m=1}^{M}\mathbf{u}_m^{\top}\mathbf{u}_m$

Algorithm 9: Selfmade - Part (d): MC Sampling

for $b = 1, \ldots, B$ **do**

`convergenceIssue` $\leftarrow$ `TRUE`

while *`convergenceIssue`* **do**

Draw $T_b \sim \mathcal{D}$ with $\mathcal{D} = \mathcal{N}(0, \kappa^2)$ or $\mathcal{D} = \mathcal{N}(t_{\text{obs}}, \kappa^2)$

Apply transformation $\mathbf{y}_b = \mathbf{dir}_{\mathbf{v}} \cdot T_b + P_v^{\perp}(\mathbf{y}_{\text{obs}})$

Apply selection procedure $\mathcal{S}(\mathbf{y}_b)$

if *convergence check is used* **then**

`convergenceIssue` $\leftarrow \mathbb{1}_{\{\text{convergence issue in } \mathcal{S}(\mathbf{y}_b) \text{ occurred}\}}$

else

`convergenceIssue` $\leftarrow$ `FALSE`

if $\mathcal{S}(\mathbf{y}_b) = \mathcal{S}(\mathbf{y}_{obs})$ **then**

$\mathbb{1}_{\{T_b \in \mathcal{T}\}} \leftarrow 1$

else if $\mathcal{S}(\mathbf{y}_b) \neq \mathcal{S}(\mathbf{y}_{obs})$ **then**

$\mathbb{1}_{\{T_b \in \mathcal{T}\}} \leftarrow 0$

if $T_b \geq t_{\text{obs}}$ **then**

$\mathbb{1}_{\{T_b \geq t_{\text{obs}}\}} \leftarrow 1$

else if $T_b < t_{\text{obs}}$ **then**

$\mathbb{1}_{\{T_b \geq t_{\text{obs}}\}} \leftarrow 0$

Compute importance sampling weight as ratio of target and proposal densities

$w_b = \frac{f_{\mathcal{N}(0,\kappa^2)}(T_b)}{f_{\mathcal{D}}(T_b)}$

Compute right-sided p-value $p_{rightsided} = \zeta(\rho) = \frac{\sum_{b=1}^{B} w_b \cdot \exp(\frac{T_b \cdot \rho}{\kappa^2}) \mathbb{1}_{\{T_b \in \mathcal{T}\}} \cdot \mathbb{1}_{\{T_b \geq t_{\text{obs}}\}}}{\sum_{b=1}^{B} w_b \cdot \exp(\frac{T_b \cdot \rho}{\kappa^2}) \mathbb{1}_{\{T_b \in \mathcal{T}\}}}$ [a]

Compute two-sided p-value $p_{twosided} = 2 \cdot \min(p_{rightsided}, 1 - p_{rightsided})$

Compute confidence interval by solving $\zeta(\rho_{\alpha/2}) = \alpha/2$ and $\zeta(\rho_{1-\alpha/2}) = 1 - \alpha/2$

[a] If no sampled value reproduces the selection event, the denominator is zero and $\zeta(\rho)$ is undefined. In practice, a minimum number of samples reproducing the selection event can be required to improve numerical stability. This was not implemented in Rügamer et al. (2022); we follow their approach here. Our implementation nevertheless allows a minimum number of accepted samples to be specified.

Equivalence between Equations (9) **and** (10)

$$
\begin{aligned}
\zeta(\rho) &= \int_{T \geq t_{\text{obs}}} f_{\mathcal{N}\text{truncated}(\rho,\kappa^2)}(T)\, dT \\
&= \frac{\int_{T \geq t_{\text{obs}}} f_{\mathcal{N}\text{truncated}(\rho,\kappa^2)}(T)\, dT}{1} \\
&= \frac{\int_{T \geq t_{\text{obs}}} f_{\mathcal{N}\text{truncated}(\rho,\kappa^2)}(T)\, dT}{\int_{\mathbb{R}} f_{\mathcal{N}\text{truncated}(\rho,\kappa^2)}(T)\, dT} \\
&\overset{(8)}{=} \frac{\int_{T \geq t_{\text{obs}}} c \cdot f_{\mathcal{N}(\rho,\kappa^2)}(T) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} dT}{\int_{\mathbb{R}} c \cdot f_{\mathcal{N}(\rho,\kappa^2)}(T) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} dT} \\
&= \frac{c \cdot \int_{\mathbb{R}} f_{\mathcal{N}(\rho,\kappa^2)}(T) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} \cdot \mathbb{1}_{\{T \geq t_{\text{obs}}\}} dT}{c \cdot \int_{\mathbb{R}} f_{\mathcal{N}(\rho,\kappa^2)}(T) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} dT} \\
&= \frac{\mathbb{E}_{T \sim \mathcal{N}(\rho,\kappa^2)} \left[\mathbb{1}_{\{T \in \mathcal{T}\}} \cdot \mathbb{1}_{\{T \geq t_{\text{obs}}\}} \right]}{\mathbb{E}_{T \sim \mathcal{N}(\rho,\kappa^2)} \left[\mathbb{1}_{\{T \in \mathcal{T}\}} \right]}
\end{aligned}
\tag{S1}
$$

□

Rewriting Equation (10)

$$
\begin{aligned}
\zeta(\rho) &= \frac{\mathbb{E}_{T \sim \mathcal{N}(\rho,\kappa^2)} \left[\mathbb{1}_{\{T \in \mathcal{T}\}} \cdot \mathbb{1}_{\{T \geq t_{\text{obs}}\}} \right]}{\mathbb{E}_{T \sim \mathcal{N}(\rho,\kappa^2)} \left[\mathbb{1}_{\{T \in \mathcal{T}\}} \right]} \\
&= \frac{\int_{\mathbb{R}} f_{\mathcal{N}(\rho,\kappa^2)}(T) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} \cdot \mathbb{1}_{\{T \geq t_{\text{obs}}\}} dT}{\int_{\mathbb{R}} f_{\mathcal{N}(\rho,\kappa^2)}(T) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} dT} \\
&= \frac{\int_{\mathbb{R}} \exp\left(-\frac{1}{2} \cdot \frac{(T-\rho)^2}{\kappa^2}\right) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} \cdot \mathbb{1}_{\{T \geq t_{\text{obs}}\}} dT}{\int_{\mathbb{R}} \exp\left(-\frac{1}{2} \cdot \frac{(T-\rho)^2}{\kappa^2}\right) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} dT} \\
&= \frac{\int_{\mathbb{R}} \exp\left(-\frac{1}{2} \cdot \frac{T^2 - 2 \cdot T \cdot \rho + \rho^2}{\kappa^2}\right) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} \cdot \mathbb{1}_{\{T \geq t_{\text{obs}}\}} dT}{\int_{\mathbb{R}} \exp\left(-\frac{1}{2} \cdot \frac{T^2 - 2 \cdot T \cdot \rho + \rho^2}{\kappa^2}\right) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} dT} \\
&= \frac{\exp\left(-\frac{1}{2} \cdot \frac{\rho^2}{\kappa^2}\right) \cdot \int_{\mathbb{R}} \exp\left(-\frac{1}{2} \cdot \frac{(T-0)^2}{\kappa^2}\right) \cdot \exp\left(\frac{T \cdot \rho}{\kappa^2}\right) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} \cdot \mathbb{1}_{\{T \geq t_{\text{obs}}\}} dT}{\exp\left(-\frac{1}{2} \cdot \frac{\rho^2}{\kappa^2}\right) \cdot \int_{\mathbb{R}} \exp\left(-\frac{1}{2} \cdot \frac{(T-0)^2}{\kappa^2}\right) \cdot \exp\left(\frac{T \cdot \rho}{\kappa^2}\right) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} dT} \\
&= \frac{\mathbb{E}_{T \sim \mathcal{N}(0,\kappa^2)} \left[\exp\left(\frac{T \cdot \rho}{\kappa^2}\right) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} \cdot \mathbb{1}_{\{T \geq t_{\text{obs}}\}} \right]}{\mathbb{E}_{T \sim \mathcal{N}(0,\kappa^2)} \left[\exp\left(\frac{T \cdot \rho}{\kappa^2}\right) \cdot \mathbb{1}_{\{T \in \mathcal{T}\}} \right]}
\end{aligned}
\tag{S2}
$$

□

Selfmade: Importance Sampling

In the Selfmade approach, the conditional sampling distribution of the test statistic is approximated by MC sampling. Specifically, samples T_b are drawn from the target distribution $\mathcal{N}(\rho, \kappa^2)$ and transformed into outcome vectors $\mathbf{y}_b$. The selection procedure is then applied to each $\mathbf{y}_b$, and only samples for which $\mathcal{S}(\mathbf{y}_b) = \mathcal{A}$ are retained. This rejection sampling procedure can become computationally inefficient when few or no sampled values fall within the inference region $\mathcal{T}$, for example, when $\mathcal{T}$ lies in a region where the target density is low.

To improve computational efficiency, Rügamer et al. (2022) proposed *importance sampling*. Instead of sampling directly from the target distribution $\mathcal{N}(\rho, \kappa^2)$, samples are drawn from a proposal distribution that places more mass in the inference region $\mathcal{T}$. Let $p(t)$ denote the density of the proposal distribution and $t_\rho(t)$ the density of the target distribution. Each sampled value T_b is then assigned the importance weight

$$w_b = \frac{t_\rho(T_b)}{p(T_b)}. \tag{S3}$$

The weights account for the difference between the proposal and target distributions and allow expectations under the target distribution to be approximated using samples from the proposal distribution.

Accordingly, the right-sided conditional p-value can be estimated as

$$\frac{\sum_{b=1}^{B} w_b \, \mathbb{1}_{\{T_b \in \mathcal{T}\}} \mathbb{1}_{\{T_b \geq t_{\text{obs}}\}}}{\sum_{b=1}^{B} w_b \, \mathbb{1}_{\{T_b \in \mathcal{T}\}}}. \tag{S4}$$

Proof of the Proposition Discussed in Simulation Study 1

Proposition 1. *Consider an LMM with a fixed intercept, one Level-2 predictor, and a random intercept for each of M Level-2 units. Let* $m = 1, \ldots, M$ *index the Level-2 units, and let* N_m *denote the number of Level-1 observations in unit m. For simplicity, assume balanced cluster sizes, so that* $N_m = N/M$ *for all m. Let*

$$\mathbf{R} = \sigma_\varepsilon^2 \mathbf{I}_N, \qquad \mathbf{G} = \tau_0^2 \mathbf{I}_M, \tag{S5}$$

with $\sigma_\varepsilon^2, \tau_0^2 > 0$ *and define*

$$\mathbf{X} = \begin{pmatrix} \mathbf{1}_{N_1} & x_1 \mathbf{1}_{N_1} \\ \mathbf{1}_{N_2} & x_2 \mathbf{1}_{N_2} \\ \vdots & \vdots \\ \mathbf{1}_{N_M} & x_M \mathbf{1}_{N_M} \end{pmatrix} \tag{S6}$$

and

$$\mathbf{Z} = \begin{pmatrix} \mathbf{1}_{N_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{1}_{N_2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{1}_{N_M} \end{pmatrix}. \tag{S7}$$

Let

$$S_{xx,c} = \sum_{m=1}^{M} (x_m - \bar{x})^2 > 0, \qquad \bar{x} = \frac{1}{M} \sum_{m=1}^{M} x_m. \tag{S8}$$

Then the frequentist and Bayesian conditional variances of the test statistic for β_1 *are*

$$\kappa^2 = \frac{\sigma_\varepsilon^2}{N_m S_{xx,c}}, \quad \kappa^2_{Bayesian} = \frac{\tau_0^2 + \sigma_\varepsilon^2 / N_m}{S_{xx,c}}. \tag{S9}$$

Consequently,

$$\kappa^2_{Bayesian} - \kappa^2 = \frac{\tau_0^2}{S_{xx,c}}. \tag{S10}$$

Proof. By definition,

$$\mathbf{A} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \tau_0^{-2} \mathbf{I}_M \end{pmatrix}. \tag{S11}$$

Because $\mathbf{R}^{-1} = \sigma_\varepsilon^{-2}\mathbf{I}_N$, we define $a = \frac{N_m}{\sigma_\varepsilon^2}$. Let

$$\mathbf{x} = (x_1, \ldots, x_M)^\top, \quad \mathbf{1} = (1, \ldots, 1)^\top. \tag{S12}$$

The cross-products required below are

$$\mathbf{X}^\top\mathbf{R}^{-1}\mathbf{X} = a\begin{pmatrix} M & \mathbf{1}^\top\mathbf{x} \\ \mathbf{1}^\top\mathbf{x} & \mathbf{x}^\top\mathbf{x} \end{pmatrix}, \tag{S13}$$

$$\mathbf{X}^\top\mathbf{R}^{-1}\mathbf{Z} = a\begin{pmatrix} \mathbf{1}^\top \\ \mathbf{x}^\top \end{pmatrix}, \tag{S14}$$

and

$$\mathbf{Z}^\top\mathbf{R}^{-1}\mathbf{Z} = a\mathbf{I}_M. \tag{S15}$$

Without loss of generality, we center the Level-2 predictor so that $\bar{x} = 0$. This reparameterization leaves the coefficient of the Level-2 predictor unchanged. It follows that

$$\mathbf{1}^\top\mathbf{x} = 0 \qquad \text{and} \qquad \mathbf{x}^\top\mathbf{x} = S_{xx,c}. \tag{S16}$$

The fixed intercept is therefore orthogonal to the Level-2 predictor. Order the coefficient vector as $(\beta_0, \beta_1, \mathbf{b}^\top)^\top$, so that $\mathbf{K} = \mathbf{C}^\top\mathbf{R}^{-1}\mathbf{C} + \mathbf{A}$ has dimension $(M+2) \times (M+2)$ and, using $\mathbf{1}^\top\mathbf{x} = 0$,

$$\mathbf{K} = \begin{pmatrix} aM & 0 & a\mathbf{1}^\top \\ 0 & aS_{xx,c} & a\mathbf{x}^\top \\ a\mathbf{1} & a\mathbf{x} & d\mathbf{I}_M \end{pmatrix}, \qquad d = a + \tau_0^{-2}. \tag{S17}$$

Let $\mathbf{e}_1$ denote the unit vector selecting β_1 and consider $\mathbf{K}\mathbf{w}^{\text{full}} = \mathbf{e}_1$, where $\mathbf{w}^{\text{full}} = (w_{\beta_0}, w_0, \mathbf{w}_b^\top)^\top$. The corresponding system of equations is

$$\begin{aligned} aMw_{\beta_0} + a\mathbf{1}^\top\mathbf{w}_b &= 0, \\ aS_{xx,c}w_0 + a\mathbf{x}^\top\mathbf{w}_b &= 1, \\ a\mathbf{1}w_{\beta_0} + a\mathbf{x}w_0 + d\mathbf{w}_b &= \mathbf{0}. \end{aligned} \tag{S18}$$

The first and third equations are satisfied by $w_{\beta_0} = 0$ and $\mathbf{w}_b = -\frac{a}{d}\mathbf{x}w_0$, since $\mathbf{1}^\top\mathbf{x} = 0$. Substituting this into the second equation yields

$$aS_{xx,c}w_0 - \frac{a^2}{d}S_{xx,c}w_0 = 1 \implies w_0 = \frac{d}{aS_{xx,c}(d-a)}. \tag{S19}$$

Since $\mathbf{K}$ is positive definite, the solution is unique. Hence, the intercept component of $\mathbf{K}^{-1}\mathbf{e}_1$ is zero, and its remaining components can be obtained from the reduced system involving

$$\mathbf{K}_{\beta_1} = \begin{pmatrix} aS_{xx,c} & a\mathbf{x}^\top \\ a\mathbf{x} & d\mathbf{I}_M \end{pmatrix} \in \mathbb{R}^{(M+1)\times(M+1)}. \tag{S20}$$

This reduction is used to derive both the Bayesian and the frequentist conditional variances below.

The Bayesian variance requires the $(1,1)$ element of $\mathbf{K}_{\beta_1}^{-1}$. By the Schur complement,

$$\kappa^2_{\text{Bayesian}} = \left(aS_{xx,c} - a\mathbf{x}^\top(d\mathbf{I}_M)^{-1}a\mathbf{x}\right)^{-1} = \left(aS_{xx,c} - \frac{a^2}{d}S_{xx,c}\right)^{-1} = \frac{d}{aS_{xx,c}(d-a)}. \tag{S21}$$

Because $d - a = \tau_0^{-2}$, we obtain

$$\kappa^2_{\text{Bayesian}} = \frac{\tau_0^2 d}{aS_{xx,c}}. \tag{S22}$$

Substituting $a = \frac{N_m}{\sigma_\varepsilon^2}$ and $d = \frac{N_m}{\sigma_\varepsilon^2} + \tau_0^{-2}$ yields

$$\kappa^2_{\text{Bayesian}} = \frac{\tau_0^2\left(\frac{N_m}{\sigma_\varepsilon^2} + \tau_0^{-2}\right)}{\frac{N_m}{\sigma_\varepsilon^2}S_{xx,c}} = \frac{\tau_0^2 + \frac{\sigma_\varepsilon^2}{N_m}}{S_{xx,c}}. \tag{S23}$$

The frequentist conditional variance is given by

$$\kappa^2 = \mathbf{w}^\top\mathbf{C}^\top\mathbf{R}^{-1}\mathbf{C}\mathbf{w}. \tag{S24}$$

Because $w_{\beta_0} = 0$, the intercept component of $\mathbf{w}^{\text{full}}$ does not contribute to the quadratic form, and $(\mathbf{w}^{\text{full}})^\top\mathbf{C}^\top\mathbf{R}^{-1}\mathbf{C}\,\mathbf{w}^{\text{full}}$ equals $\mathbf{w}^\top\mathbf{M}_{\beta_1}\mathbf{w}$ with $\mathbf{w} = (w_0, \mathbf{w}_b^\top)^\top$ and the reduced

$(M+1)\times(M+1)$ matrix

$$\mathbf{M}_{\beta_1} := a\begin{pmatrix} S_{xx,c} & \mathbf{x}^\top \\ \mathbf{x} & \mathbf{I}_M \end{pmatrix} \tag{S25}$$

(the $(\beta_1, \mathbf{b})$ block of $\mathbf{C}^\top\mathbf{R}^{-1}\mathbf{C}$). We obtain

$$\begin{aligned}
\kappa^2 = \mathbf{w}^\top\mathbf{M}_{\beta_1}\mathbf{w} = \mathbf{w}^\top a\begin{pmatrix} S_{xx,c} & \mathbf{x}^\top \\ \mathbf{x} & \mathbf{I}_M \end{pmatrix}\mathbf{w} =& \mathbf{w}^\top\begin{pmatrix} S_{xx,c}w_0(1-\frac{a}{d}) \\ \mathbf{x}w_0(1-\frac{a}{d}) \end{pmatrix}a \\
=& \mathbf{w}^\top\begin{pmatrix} S_{xx,c} \\ \mathbf{x} \end{pmatrix}w_0(1-\frac{a}{d})a \\
=& w_0^2 S_{xx,c}\left(1-\frac{a}{d}\right)^2 a \\
=& a\left[\frac{d}{aS_{xx,c}(d-a)}\right]^2 S_{xx,c}\left[\frac{d-a}{d}\right]^2 \\
=& \frac{1}{aS_{xx,c}} = \frac{\sigma_\varepsilon^2}{N_m S_{xx,c}}.
\end{aligned} \tag{S26}$$

Finally, subtracting the frequentist conditional variance from the Bayesian conditional variance gives

$$\kappa^2_{\text{Bayesian}} - \kappa^2 = \frac{\tau_0^2 + \frac{\sigma_\varepsilon^2}{N_m}}{S_{xx,c}} - \frac{\sigma_\varepsilon^2}{N_m S_{xx,c}} = \frac{\tau_0^2}{S_{xx,c}}. \tag{S27}$$

□

Inference Targets under the Marginal and Bayesian Conditional Perspective, and Why the Conditional Perspective Fails to Provide Valid Inference in Simulation Study 3

Simulation Studies 2–4 share the same data-generating mechanism. We derive the inference targets under our simulation design explicitly for the full linear working model $\mathbf{X}_{\mathcal{A}} = (\mathbf{1}_N, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4)$. The same derivation applies to the working model $\mathbf{X}_{\mathcal{A}} = (\mathbf{1}_N, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_4)$, yielding an inference target of zero for x_4. For the working model $\mathbf{X}_{\mathcal{A}} = (\mathbf{1}_N, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$, the inference target for x_3 is likewise zero. This follows from the construction and orthogonality of the predictors and the absence of a random slope for x_3 in the data-generating model. We omit the corresponding derivation for brevity.

Because the predictor matrix is replicated identically across all $M = 40$ Level-2 units, it is convenient to introduce the common *within-cluster template*

$$\mathbf{X}_{\mathcal{A},0} = (\mathbf{1}_{N_m}, \mathbf{x}_{1,m}, \mathbf{x}_{2,m}, \mathbf{x}_{3,m}, \mathbf{x}_{4,m}) \quad (N_m \times 5), \qquad \mathbf{X}_{\mathcal{A}} = \mathbf{1}_M \otimes \mathbf{X}_{\mathcal{A},0} \quad (N \times 5), \tag{S28}$$

that is, $\mathbf{X}_{\mathcal{A},0}$ is the identical within-cluster design shared by every unit m, and $\mathbf{X}_{\mathcal{A}}$ stacks M copies of it. Analogously, the true random-effects design is

$$\mathbf{Z} = \mathbf{I}_M \otimes \mathbf{Z}_0 \quad (N \times 3M), \qquad \mathbf{Z}_0 = (\mathbf{1}_{N_m}, \mathbf{x}_{2,m}, \mathbf{x}_{4,m}) \quad (N_m \times 3), \tag{S29}$$

that is, $\mathbf{Z}$ is block-diagonal with M identical $N_m \times 3$ blocks $\mathbf{Z}_0 = \mathbf{Z}_m$. With $\mathbf{G} = \mathbf{I}_M \otimes \mathbf{D}$ (i.i.d draws of $\mathbf{b}_m \sim \mathcal{N}(\mathbf{0}, \mathbf{D})$ across clusters) and $\mathbf{R} = \sigma_\varepsilon^2 \mathbf{I}_N$,

$$\boldsymbol{\mu}_{\text{FE}} = \mathbf{1} - 0.5\,\mathbf{x}_1^2 + 0.5\,\mathbf{x}_2^2 - 4\,\mathbf{x}_3 + \mathbf{x}_3^2 + 4\,\mathbf{x}_4 - \mathbf{x}_4^2, \quad \boldsymbol{\Sigma} = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R} = \mathbf{I}_M \otimes \boldsymbol{\Sigma}_0, \tag{S30}$$

where $\boldsymbol{\Sigma}_0 := \mathbf{Z}_0 \mathbf{D} \mathbf{Z}_0^\top + \sigma_\varepsilon^2 \mathbf{I}_{N_m}$ is the identical within-cluster covariance template; $\boldsymbol{\Sigma}$ itself is block-diagonal with M identical $N_m \times N_m$ blocks. The studies differ only in the random-effects structure of the working model:

- *Simulation Study 2:* $\mathbf{Z}_{\mathcal{B}} = \mathbf{I}_M \otimes \mathbf{Z}_0 = \mathbf{Z}$, with $\mathbf{G}_{\text{working}} \approx \mathbf{G}$.
- *Simulation Study 3:* $\mathbf{Z}_{\mathcal{B}} = \mathbf{I}_M \otimes \mathbf{Z}_{0,\mathcal{B}}$ with $\mathbf{Z}_{0,\mathcal{B}} = \mathbf{1}_{N_m}$, i.e. only a random intercept per cluster.

- *Simulation Study 4:* $\mathbf{Z}_{\mathcal{B}}$ depends on selected random effects.

New random effects $\mathbf{b}$ are redrawn independently in every simulation replication.

This section shows that Simulation Studies 2–4 yield identical unconditional inference targets for each perspective considered, for both x_3 and x_4, and that the difference in Type-I error behavior reported for β_4 between Simulation Studies 2 and 3 does not originate from the target of inference under the Bayesian conditional perspective itself, but from inadequate uncertainty quantification caused by the misspecified random-effects structure under our simulation design.

Equivalence of the Marginal Inferential Targets

For the two marginal perspectives, the targets are $\mathbf{v}^\top \boldsymbol{\mu}_{\mathrm{FE}}$ and $\mathbf{v}_{\mathrm{eff}}^\top \boldsymbol{\mu}_{\mathrm{FE}}$, with $\mathbf{v}$ and $\mathbf{v}_{\mathrm{eff}}$ computed from $\mathbf{X}_{\mathcal{A}}$ and the true $\boldsymbol{\Sigma}$ (see Equations (16)–(19)). Neither target involves $\mathbf{Z}_{\mathcal{B}}$ or $\mathbf{G}_{\mathrm{working}}$. Consequently, the marginal targets cannot differ between Simulation Study 2, Simulation Study 3 and Simulation Study 4.

Because $\boldsymbol{\mu}_{\mathrm{FE}} \neq \mathbf{X}_{\mathcal{A}}\boldsymbol{\beta}$ (the working model omits the quadratic terms present in $\boldsymbol{\mu}_{\mathrm{FE}}$), the two marginal targets need not coincide with each other in general (see footnote 5, which establishes coincidence only under first-order correctness). In the present design they coincide exactly, however, by a distinct mechanism: GLS equals OLS as estimators (Zyskind, 1967), whenever

$$\mathrm{col}(\boldsymbol{\Sigma}\mathbf{X}_{\mathcal{A}}) \subseteq \mathrm{col}(\mathbf{X}_{\mathcal{A}})\,. \tag{S31}$$

Since $\boldsymbol{\Sigma} = \mathbf{I}_M \otimes \boldsymbol{\Sigma}_0$ and $\mathbf{X}_{\mathcal{A}} = \mathbf{1}_M \otimes \mathbf{X}_{\mathcal{A},0}$ ((S28)–(S30)),

$$\boldsymbol{\Sigma}\mathbf{X}_{\mathcal{A}} = \mathbf{1}_M \otimes \left(\boldsymbol{\Sigma}_0\mathbf{X}_{\mathcal{A},0}\right), \tag{S32}$$

so (S31) reduces to the analogous column-space condition for the within-cluster templates,

$$\mathrm{col}(\boldsymbol{\Sigma}_0\mathbf{X}_{\mathcal{A},0}) \subseteq \mathrm{col}(\mathbf{X}_{\mathcal{A},0}). \tag{S33}$$

Since $\mathbf{Z}_0 = (\mathbf{1}_{N_m}, \mathbf{x}_{2,m}, \mathbf{x}_{4,m})$ is a genuine submatrix of $\mathbf{X}_{\mathcal{A},0} = (\mathbf{1}_{N_m}, \mathbf{x}_{1,m}, \mathbf{x}_{2,m}, \mathbf{x}_{3,m}, \mathbf{x}_{4,m})$,

there exists a 5×3 selection matrix $\mathbf{S}$, such that $\mathbf{Z}_0 = \mathbf{X}_{\mathcal{A},0}\mathbf{S}$. With that,

$$\boldsymbol{\Sigma}_0\mathbf{X}_{\mathcal{A},0} = \left(\mathbf{Z}_0\mathbf{D}\mathbf{Z}_0^\top + \sigma_\varepsilon^2\mathbf{I}_{N_m}\right)\mathbf{X}_{\mathcal{A},0} = \mathbf{X}_{\mathcal{A},0}\left(\mathbf{S}\mathbf{D}\mathbf{S}^\top\mathbf{X}_{\mathcal{A},0}^\top\mathbf{X}_{\mathcal{A},0} + \sigma_\varepsilon^2\mathbf{I}\right) \in \mathrm{col}\left(\mathbf{X}_{\mathcal{A},0}\right). \tag{S34}$$

Hence $\hat{\boldsymbol{\beta}}_{\mathcal{A},\mathrm{GLS}} = \hat{\boldsymbol{\beta}}_{\mathcal{A},\mathrm{OLS}}$ exactly, as random variables, and thus

$$\mathbf{v}^\top\boldsymbol{\mu}_{\mathrm{FE}} = \mathbf{v}_{\mathrm{eff}}^\top\boldsymbol{\mu}_{\mathrm{FE}} \tag{S35}$$

exactly. This is a special property of the present, nested and replicated random-effects design, not a general fact about misspecified working models.

Writing $\mathbf{x}_{p,m} = 2 \cdot \mathbf{1}_{N_m} + \mathbf{z}_{p,m}$ for each cluster m with $\mathbf{1}_{N_m}^\top\mathbf{z}_{p,m} = 0$, the mean function can be expressed as

$$\boldsymbol{\mu}_{\mathrm{FE}} = \mathbf{1} - 2\mathbf{z}_1 - 0.5\mathbf{z}_1^2 + 2\mathbf{z}_2 + 0.5\mathbf{z}_2^2 + \mathbf{z}_3^2 - \mathbf{z}_4^2. \tag{S36}$$

By construction of the predictors using the reflect-then-whiten procedure, $\mathbf{z}_p^\top\mathbf{z}_p^2 = 0$, and the linear and quadratic terms are orthogonal to one another, as are the distinct linear terms. Hence, for the linear working model (omitting the quadratic terms), the OLS projection coefficient for predictor z_p is

$$\mathbf{v}^\top\boldsymbol{\mu}_{\mathrm{FE}} = (\mathbf{z}_p^\top\mathbf{z}_p)^{-1}\mathbf{z}_p^\top\boldsymbol{\mu}_{\mathrm{FE}} = \begin{cases} 0, & p = 3, \\ 0, & p = 4. \end{cases} \tag{S37}$$

Since the shift from $\mathbf{x}_p$ to $\mathbf{z}_p$ only changes the intercept and leaves the slope coefficients unchanged (S37) also gives the OLS working-model targets for the original predictors x_p. Together with (S35) this confirms $\rho_3 = \rho_4 = 0$ for both marginal perspectives, identically in Simulation Study 2, Simulation Study 3 and Simulation Study 4.

Inference Targets under the Bayesian Conditional Perspective

The Bayesian conditional perspective conditions on the realized random effects. To characterize the corresponding inference target, it is useful to distinguish the inferential target from the conditional expectation of the penalized test statistic. For a working model with

fixed-effects design $\mathbf{X}_{\mathcal{A}}$ and random-effects design $\mathbf{Z}_{\mathcal{B}}$, let

$$\mathbf{C} = (\mathbf{X}_{\mathcal{A}} \;\; \mathbf{Z}_{\mathcal{B}}), \qquad \mathbf{K} = \mathbf{C}^{\top}\mathbf{R}^{-1}\mathbf{C} + \mathbf{A}, \tag{S38}$$

where

$$\mathbf{A} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{G}_{\text{working}}^{-1} \end{pmatrix}. \tag{S39}$$

The fixed-effects component of the solution to the Henderson mixed-model equations can be written as a GLS estimator based on the working marginal covariance

$$\mathbf{\Sigma}_{\text{working}} = \mathbf{Z}_{\mathcal{B}}\mathbf{G}_{\text{working}}\mathbf{Z}_{\mathcal{B}}^{\top} + \mathbf{R}, \tag{S40}$$

such that

$$\widehat{\boldsymbol{\beta}}_{\text{conditional}} = \left(\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{\Sigma}_{\text{working}}^{-1}\mathbf{X}_{\mathcal{A}}\right)^{-1}\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{\Sigma}_{\text{working}}^{-1}\mathbf{y}. \tag{S41}$$

This follows from the Henderson mixed-model equations (Robinson, 1991). In the present simulation design, the working covariance satisfies the Zyskind condition for both random-effects specifications considered. Because the fixed-effects design is replicated identically across clusters, $\mathbf{X}_{\mathcal{A}} = \mathbf{1}_M \otimes \mathbf{X}_{\mathcal{A},0}$, and the working random-effects design has the block-diagonal form $\mathbf{Z}_{\mathcal{B}} = \mathbf{I}_M \otimes \mathbf{Z}_{\mathcal{B},0}$, the working covariance can be written as

$$\mathbf{\Sigma}_{\text{working}} = \mathbf{I}_M \otimes \mathbf{\Sigma}_{0,\text{working}}, \qquad \mathbf{\Sigma}_{0,\text{working}} = \mathbf{Z}_{\mathcal{B},0}\mathbf{D}_{\text{working}}\mathbf{Z}_{\mathcal{B},0}^{\top} + \sigma_{\varepsilon}^{2}\mathbf{I}_{N_m}. \tag{S42}$$

Therefore,

$$\mathbf{\Sigma}_{\text{working}}\mathbf{X}_{\mathcal{A}} = \mathbf{1}_M \otimes \left(\mathbf{\Sigma}_{0,\text{working}}\mathbf{X}_{\mathcal{A},0}\right), \tag{S43}$$

so that the Zyskind condition reduces to the corresponding within-cluster condition

$$\text{col}\left(\mathbf{\Sigma}_{0,\text{working}}\mathbf{X}_{\mathcal{A},0}\right) \subseteq \text{col}\left(\mathbf{X}_{\mathcal{A},0}\right). \tag{S44}$$

In Simulation Study 2, the working random-effects design

$$\mathbf{Z}_{\mathcal{B},0} = (\mathbf{1}_{N_m}, \mathbf{x}_2, \mathbf{x}_4) \tag{S45}$$

is a submatrix of

$$\mathbf{X}_{\mathcal{A},0} = (\mathbf{1}_{N_m}, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4). \tag{S46}$$

Hence, there exists a 5×3 selection matrix $\mathbf{S}$ such that

$$\mathbf{Z}_{\mathcal{B},0} = \mathbf{X}_{\mathcal{A},0}\mathbf{S}, \tag{S47}$$

and therefore

$$\begin{aligned} \boldsymbol{\Sigma}_{0,\text{working}}\mathbf{X}_{\mathcal{A},0} &= \left(\mathbf{Z}_{\mathcal{B},0}\mathbf{D}_{\text{working}}\mathbf{Z}_{\mathcal{B},0}^{\top} + \sigma_{\varepsilon}^2\mathbf{I}_{N_m}\right)\mathbf{X}_{\mathcal{A},0} \\ &= \mathbf{X}_{\mathcal{A},0}\left(\mathbf{S}\mathbf{D}_{\text{working}}\mathbf{S}^{\top}\mathbf{X}_{\mathcal{A},0}^{\top}\mathbf{X}_{\mathcal{A},0} + \sigma_{\varepsilon}^2\mathbf{I}_5\right), \end{aligned} \tag{S48}$$

which lies in $\text{col}(\mathbf{X}_{\mathcal{A},0})$. Note that this argument holds for *any* $\mathbf{D}_{\text{working}} \succ 0$; the Zyskind/Henderson equivalence of the point estimator does not require $\mathbf{D}_{\text{working}} = \mathbf{D}$.

In Simulation Study 3, the working random-effects design contains only the random intercept,

$$\mathbf{Z}_{\mathcal{B},0} = \mathbf{1}_{N_m}, \tag{S49}$$

which is itself a column of $\mathbf{X}_{\mathcal{A},0}$. Hence, the Zyskind condition is satisfied under both working random-effects specifications. Consequently, the GLS estimator based on $\boldsymbol{\Sigma}_{\text{working}}$ (and thus the conditional estimator) coincides exactly with the OLS estimator in both Simulation Study 2 and Simulation Study 3. Importantly, this equivalence concerns the point estimator only. The working covariance, and therefore the uncertainty assigned to the estimator through $\kappa^2_{\text{Bayesian}}$, still differs between the two studies.

The inferential target under the conditional perspective is the corresponding coefficient of the unpenalized working-model projection,

$$\breve{\rho}_{p_j} = \mathbf{e}_j^{\top}\breve{\mathbf{V}}\boldsymbol{\mu}(\mathbf{b}), \qquad \breve{\mathbf{V}} = \left(\mathbf{C}^{\top}\mathbf{R}^{-1}\mathbf{C}\right)^{-1}\mathbf{C}^{\top}\mathbf{R}^{-1}, \tag{S50}$$

whenever the corresponding coefficient is identifiable. For non-identifiable directions, the resulting coefficient depends on the chosen resolution of the non-identifiability. In contrast, the conditional expectation of the penalized test statistic is

$$\rho_{p_j}(\mathbf{b}) = \mathbb{E}[T_{p_j} \mid \mathbf{b}] = \mathbf{e}_j^\top \mathbf{V}\boldsymbol{\mu}(\mathbf{b}), \qquad \mathbf{V} = \mathbf{K}^{-1}\mathbf{C}^\top\mathbf{R}^{-1}. \tag{S51}$$

Thus, $\breve{\rho}_{p_j}$ denotes the inferential target, whereas $\rho_{p_j}(\mathbf{b})$ denotes the conditional center of the penalized test statistic.[1] Because $\mathbf{V}$ incorporates the penalty through $\mathbf{G}_{\text{working}}$, the two quantities need not coincide in general when shrinkage is present.

For Simulation Studies 2 and 3, the fixed-effects design and the data-generating mean function are identical. Writing, within cluster m,

$$\mathbf{x}_{p,m} = 2 \cdot \mathbf{1}_{N_m} + \mathbf{z}_{p,m}, \qquad \mathbf{1}_{N_m}^\top \mathbf{z}_{p,m} = 0, \tag{S52}$$

the fixed-effects component of the data-generating mean can be written as

$$\boldsymbol{\mu}_{\text{FE}} = \mathbf{1} - 2\mathbf{z}_1 - 0.5\mathbf{z}_1^2 + 2\mathbf{z}_2 + 0.5\mathbf{z}_2^2 + \mathbf{z}_3^2 - \mathbf{z}_4^2. \tag{S53}$$

Hence, the quadratic fixed-effects component induces no linear contribution for either x_3 or x_4.

For x_4, the DGM contains a cluster-specific random slope. In cluster m, the relevant part of the conditional mean is

$$4 \cdot \mathbf{x}_{4,m} - \mathbf{x}_{4,m}^2 + b_{4m}\mathbf{x}_{4,m} = 4 \cdot \mathbf{1}_{N_m} - \mathbf{z}_{4,m}^2 + 2 \cdot b_{4m} \cdot \mathbf{1}_{N_m} + b_{4m}\mathbf{z}_{4,m}. \tag{S54}$$

The fixed-effects component therefore contributes no linear term in $\mathbf{z}_{4,m}$, whereas the random slope contributes the linear term $b_{4m}\mathbf{z}_{4,m}$. By the Henderson equivalence and the Zyskind condition established above, the fixed-effects part of $\mathbf{V}$ is identical to the OLS operator in the present design. Hence,

$$\rho_4(\mathbf{b}) = (\mathbf{z}_4^\top\mathbf{z}_4)^{-1}\mathbf{z}_4^\top\boldsymbol{\mu}(\mathbf{b}). \tag{S55}$$

[1] we slightly abuse notation here to make clear that $\breve{\rho}_{p_j}, \rho_{p_j}$ correspond to predictor p

Because the predictor values are identical across clusters, $\mathbf{z}_{4,m}$ can be written as $\mathbf{z}_{4,0}$ for every m, and therefore

$$\begin{aligned}\rho_4(\mathbf{b}) &= \frac{\sum_{m=1}^{M} \mathbf{z}_{4,m}^{\top} \boldsymbol{\mu}_m(\mathbf{b})}{\sum_{m=1}^{M} \mathbf{z}_{4,m}^{\top} \mathbf{z}_{4,m}} \\ &= \frac{\sum_{m=1}^{M} b_{4m} \mathbf{z}_{4,m}^{\top} \mathbf{z}_{4,m}}{\sum_{m=1}^{M} \mathbf{z}_{4,m}^{\top} \mathbf{z}_{4,m}} \\ &= \frac{\mathbf{z}_{4,0}^{\top} \mathbf{z}_{4,0} \sum_{m=1}^{M} b_{4m}}{M \mathbf{z}_{4,0}^{\top} \mathbf{z}_{4,0}} \\ &= \frac{1}{M} \sum_{m=1}^{M} b_{4m} =: \bar{b}_{4,M},\end{aligned} \tag{S56}$$

where the second equality follows from the orthogonality of $\mathbf{z}_{4,m}$ to the remaining terms in the conditional mean, and the final equalities follow from the identical within-cluster predictor vectors and equal cluster sizes.

For x_3, the DGM contains no random slope. Therefore, conditional on the realized random effects, no additional term proportional to x_3 is present. Together with the orthogonality induced by the reflect-then-whiten construction, this yields

$$\breve{\rho}_3 = \rho_3(\mathbf{b}) = 0. \tag{S57}$$

Thus, the inferential target for x_3 is zero in both Simulation Studies 2 and 3.

In Simulation Study 3, the x_4 random slope is absent from the working random-effects design. Hence, the x_4 direction is identifiable in the unpenalized working model, and, because $\mathbf{z}_{4,m}$ is orthogonal to the intercept and the remaining working-model predictors, the x_4 coefficient is unaffected by the random-intercept penalty. Thus,

$$\breve{\rho}_4(\mathbf{b}) = \rho_4(\mathbf{b}) = \bar{b}_{4,M}. \tag{S58}$$

Thus, conditional on a particular realization of the random effects, the inferential target is generally nonzero. Since the random effects are redrawn independently in every MC

replication from a mean-zero distribution,

$$\mathbb{E}_{\mathbf{b}}\left[\rho_4(\mathbf{b})\right] = \mathbb{E}_{\mathbf{b}}\left[\breve{\rho}_4(\mathbf{b})\right] = \mathbb{E}_{\mathbf{b}}\left[\bar{b}_{4,M}\right] = 0. \tag{S59}$$

Hence, under both Simulation Study 2 and Simulation Study 3, the corresponding marginal inferential target is zero[2], although the realization-specific conditional targets differ in their precise representation because of the non-identifiability of the x_4 fixed effect in Simulation Study 2.

Exact Calibration of the Bayesian Variance

Having established that T_4 is identical, as a random variable, across Simulation Studies 2 and 3, we now show that $\kappa^2_{\text{Bayesian}}$ differs between the two studies, and derive both values in closed form.

A single-cluster reduction. Applying the Woodbury identity to $\boldsymbol{\Sigma}^{-1}_{\text{working}} = (\mathbf{R} + \mathbf{Z}_{\mathcal{B}}\mathbf{G}_{\text{working}}\mathbf{Z}_{\mathcal{B}}^{\top})^{-1}$ shows that the fixed-effects block of $\mathbf{K}^{-1}$ satisfies

$$\left[\mathbf{K}^{-1}\right]_{\text{FE}} = \left(\mathbf{X}_{\mathcal{A}}^{\top}\boldsymbol{\Sigma}^{-1}_{\text{working}}\mathbf{X}_{\mathcal{A}}\right)^{-1}, \text{ hence } \kappa^2_{\text{Bayesian},j} = \left[\left(\mathbf{X}_{\mathcal{A}}^{\top}\boldsymbol{\Sigma}^{-1}_{\text{working}}\mathbf{X}_{\mathcal{A}}\right)^{-1}\right]_{jj}. \tag{S60}$$

Using $\mathbf{X}_{\mathcal{A}} = \mathbf{1}_M \otimes \mathbf{X}_{\mathcal{A},0}$ and $\boldsymbol{\Sigma}_{\text{working}} = \mathbf{I}_M \otimes \boldsymbol{\Sigma}_{0,\text{working}}$,

$$\mathbf{X}_{\mathcal{A}}^{\top}\boldsymbol{\Sigma}^{-1}_{\text{working}}\mathbf{X}_{\mathcal{A}} = M \cdot \mathbf{X}_{\mathcal{A},0}^{\top}\boldsymbol{\Sigma}^{-1}_{0,\text{working}}\mathbf{X}_{\mathcal{A},0}, \tag{S61}$$

---

[2] In Simulation Study 2, the x_4 fixed-effect direction is not separately identifiable from the cluster-specific x_4 random-slope directions because the same predictor vector is replicated across clusters. Consequently, $\mathbf{C}^{\top}\mathbf{R}^{-1}\mathbf{C}$ is singular in this direction, and the unpenalized target is not uniquely defined. To obtain a specific coefficient from the Moore–Penrose pseudoinverse, consider the minimum-norm solution: for each cluster, only the sum of the fixed and random slope coefficients, $\beta_4 + b_{4m}$, is identified, so the minimum-norm resolution minimizes $\beta_4^2 + \sum_m b_{4m}^2$ subject to these M constraints. Minimizing this expression yields

$$\breve{\rho}_4(\mathbf{b}) = \frac{M}{M+1}\bar{b}_{4,M}.$$

Thus, although the realization-specific target depends on the chosen resolution of the non-identifiability, its expectation over the mean-zero random-effects distribution is

$$\mathbb{E}_{\mathbf{b}}\left[\breve{\rho}_4(\mathbf{b})\right] = \frac{M}{M+1}\mathbb{E}_{\mathbf{b}}\left[\bar{b}_{4,M}\right] = 0 = \mathbb{E}_{\mathbf{b}}\left[\rho_4(\mathbf{b})\right].$$

which implies

$$\kappa^2_{\text{Bayesian},j} = \frac{1}{M}\left[\left(\mathbf{X}_{\mathcal{A},0}^\top \boldsymbol{\Sigma}_{0,\text{working}}^{-1}\mathbf{X}_{\mathcal{A},0}\right)^{-1}\right]_{jj}. \tag{S62}$$

This reduces the computation of $\kappa^2_{\text{Bayesian}}$ from an N-dimensional to a single within-cluster ($N_m \times N_m$, effectively 5×5 after taking cross-products) problem. It is convenient to work in the orthogonal within-cluster z-basis $\mathbf{X}^z_{\mathcal{A},0} = (\mathbf{1}_{N_m}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{z}_4)$, which spans the same column space as $\mathbf{X}_{\mathcal{A},0}$ and hence leaves (S62) unchanged. The intercept column is exact, $\mathbf{1}_{N_m}^\top \mathbf{1}_{N_m} = N_m$. For the predictor columns, standardization was carried out using the sample standard deviation computed across all $N = MN_m$ observations. Since the predictor pattern is replicated identically across the M clusters,

$$\mathbf{z}_{p,m}^\top \mathbf{z}_{p,m} = \frac{N-1}{M} =: c, \qquad p = 1, \ldots, 4. \tag{S63}$$

Together with the pairwise orthogonality of $\mathbf{1}_{N_m}, \mathbf{z}_1, \ldots, \mathbf{z}_4$ established above,

$$\mathbf{X}^{z\top}_{\mathcal{A},0}\mathbf{X}^z_{\mathcal{A},0} = \boldsymbol{\Lambda} := \text{diag}(N_m,\ c,\ c,\ c,\ c)\,, \qquad c = \frac{N-1}{M} = N_m - \frac{1}{M}. \tag{S64}$$

Simulation Study 2. $\mathbf{Z}_{\mathcal{B},0} = \mathbf{Z}_0$ and $\mathbf{D}_{\text{working}} = \mathbf{D}$, hence

$$\boldsymbol{\Sigma}_{0,\text{working}} = \boldsymbol{\Sigma}_0 = \mathbf{Z}_0\mathbf{D}\mathbf{Z}_0^\top + \sigma^2_\varepsilon \mathbf{I}_{N_m}, \tag{S65}$$

that is, the working covariance coincides with the true within-cluster covariance. By (S35), the resulting GLS estimator of the fixed effects is identical to the OLS estimator. Hence, its sampling variance, which equals $\kappa^2_{\text{Bayesian}}$ by (S60), can be obtained directly from

$$\text{var}\left(\widehat{\boldsymbol{\beta}}_{\mathcal{A},\text{OLS}}\right) = \left(\mathbf{X}_{\mathcal{A}}^\top\mathbf{X}_{\mathcal{A}}\right)^{-1}\mathbf{X}_{\mathcal{A}}^\top\boldsymbol{\Sigma}\mathbf{X}_{\mathcal{A}}\left(\mathbf{X}_{\mathcal{A}}^\top\mathbf{X}_{\mathcal{A}}\right)^{-1}. \tag{S66}$$

To express this variance in the orthogonal z-basis, write

$$\mathbf{Z}_0 = \mathbf{X}^z_{\mathcal{A},0}\mathbf{S}_z, \qquad \mathbf{S}_z = \mathbf{S} + \mathbf{e}_1 \cdot (0, 2, 2), \tag{S67}$$

where $\mathbf{S}$ is the selection matrix defined in (S34). The adjustment accounts for $x_2 = z_2 + 2$ and $x_4 = z_4 + 2$. By construction,

$$\mathbf{X}_{\mathcal{A},0}^{z\top}\mathbf{Z}_0 = \mathbf{X}_{\mathcal{A},0}^{z\top}\mathbf{X}_{\mathcal{A},0}^{z}\mathbf{S}_z = \mathbf{\Lambda}\mathbf{S}_z. \tag{S68}$$

It follows that

$$\mathbf{X}_{\mathcal{A},0}^{z\top}\mathbf{\Sigma}_0\mathbf{X}_{\mathcal{A},0}^{z} = \mathbf{X}_{\mathcal{A},0}^{z\top}\mathbf{Z}_0\mathbf{D}\mathbf{Z}_0^{\top}\mathbf{X}_{\mathcal{A},0}^{z} + \sigma_\varepsilon^2\mathbf{X}_{\mathcal{A},0}^{z\top}\mathbf{X}_{\mathcal{A},0}^{z} = \mathbf{\Lambda}\mathbf{S}_z\mathbf{D}\mathbf{S}_z^{\top}\mathbf{\Lambda} + \sigma_\varepsilon^2\mathbf{\Lambda}.$$

Because the same within-cluster design is replicated across the M clusters,

$$\mathbf{X}_{\mathcal{A}}^{z\top}\mathbf{X}_{\mathcal{A}}^{z} = M\mathbf{\Lambda} \tag{S69}$$

and

$$\mathbf{X}_{\mathcal{A}}^{z\top}\mathbf{\Sigma}\mathbf{X}_{\mathcal{A}}^{z} = M\left(\mathbf{\Lambda}\mathbf{S}_z\mathbf{D}\mathbf{S}_z^{\top}\mathbf{\Lambda} + \sigma_\varepsilon^2\mathbf{\Lambda}\right). \tag{S70}$$

Consequently,

$$\begin{aligned}\operatorname{var}\left(\widehat{\boldsymbol{\beta}}_{\mathcal{A},\mathrm{OLS}}^{z}\right) &= (M\mathbf{\Lambda})^{-1}M\left(\mathbf{\Lambda}\mathbf{S}_z\mathbf{D}\mathbf{S}_z^{\top}\mathbf{\Lambda} + \sigma_\varepsilon^2\mathbf{\Lambda}\right)(M\mathbf{\Lambda})^{-1} \\ &= \frac{1}{M}\mathbf{S}_z\mathbf{D}\mathbf{S}_z^{\top} + \frac{\sigma_\varepsilon^2}{M}\mathbf{\Lambda}^{-1}.\end{aligned} \tag{S71}$$

For the x_4 slope, the corresponding diagonal element of $\mathbf{S}_z\mathbf{D}\mathbf{S}_z^{\top}$ remains τ_4^2, because the shift from z_4 to $x_4 = z_4 + 2$ changes only the intercept component. Moreover,

$$\left[\mathbf{\Lambda}^{-1}\right]_{z_4,z_4} = \frac{1}{c}. \tag{S72}$$

Since the coefficient of x_4 is identical to the coefficient of z_4, its sampling variance is therefore

$$\kappa^2_{\text{Bayesian}}(\text{Study 2}, x_4) = \frac{\tau_4^2}{M} + \frac{\sigma_\varepsilon^2}{Mc} = \frac{\tau_4^2}{M} + \frac{\sigma_\varepsilon^2}{N-1}. \tag{S73}$$

The first term represents the contribution of the x_4 random slope, and the second term represents the contribution of the level-1 residual variation.

Simulation Study 3. In Simulation Study 3, the working random-effects structure contains only a random intercept,

$$\mathbf{Z}_{\mathcal{B},0} = \mathbf{1}_{N_m}, \tag{S74}$$

with

$$\mathbf{D}_{\text{working}} = \tau^2_{0,\text{working}}. \tag{S75}$$

Hence, the working within-cluster covariance matrix is

$$\boldsymbol{\Sigma}_{0,\text{working}} = \sigma^2_\varepsilon \mathbf{I}_{N_m} + \tau^2_{0,\text{working}} \mathbf{1}_{N_m} \mathbf{1}^\top_{N_m}. \tag{S76}$$

The Woodbury identity yields

$$\boldsymbol{\Sigma}^{-1}_{0,\text{working}} = \frac{1}{\sigma^2_\varepsilon} \mathbf{I}_{N_m} - \frac{1}{\sigma^4_\varepsilon} \left(\frac{N_m}{\sigma^2_\varepsilon} + \frac{1}{\tau^2_{0,\text{working}}} \right)^{-1} \mathbf{1}_{N_m} \mathbf{1}^\top_{N_m}. \tag{S77}$$

We next premultiply and postmultiply (S77) by $\mathbf{X}^z_{\mathcal{A},0}$:

$$\mathbf{X}^{z\top}_{\mathcal{A},0} \boldsymbol{\Sigma}^{-1}_{0,\text{working}} \mathbf{X}^z_{\mathcal{A},0} = \frac{1}{\sigma^2_\varepsilon} \mathbf{X}^{z\top}_{\mathcal{A},0} \mathbf{X}^z_{\mathcal{A},0} - \frac{1}{\sigma^4_\varepsilon} \left(\frac{N_m}{\sigma^2_\varepsilon} + \frac{1}{\tau^2_{0,\text{working}}} \right)^{-1} \mathbf{X}^{z\top}_{\mathcal{A},0} \mathbf{1}_{N_m} \mathbf{1}^\top_{N_m} \mathbf{X}^z_{\mathcal{A},0}.$$

By the orthogonality of the z-basis,

$$\mathbf{X}^{z\top}_{\mathcal{A},0} \mathbf{1}_{N_m} = N_m \mathbf{e}_1. \tag{S78}$$

Hence,

$$\mathbf{X}^{z\top}_{\mathcal{A},0} \mathbf{1}_{N_m} \mathbf{1}^\top_{N_m} \mathbf{X}^z_{\mathcal{A},0} = N^2_m \mathbf{e}_1 \mathbf{e}^\top_1. \tag{S79}$$

Using

$$\mathbf{X}^{z\top}_{\mathcal{A},0} \mathbf{X}^z_{\mathcal{A},0} = \boldsymbol{\Lambda}, \tag{S80}$$

we therefore obtain

$$\mathbf{X}^{z\top}_{\mathcal{A},0} \boldsymbol{\Sigma}^{-1}_{0,\text{working}} \mathbf{X}^z_{\mathcal{A},0} = \frac{1}{\sigma^2_\varepsilon} \boldsymbol{\Lambda} - \frac{N^2_m}{\sigma^4_\varepsilon} \left(\frac{N_m}{\sigma^2_\varepsilon} + \frac{1}{\tau^2_{0,\text{working}}} \right)^{-1} \mathbf{e}_1 \mathbf{e}^\top_1. \tag{S81}$$

Thus, the second term (correction term) affects only the intercept coordinate and is zero in the x_4 coordinate. In particular, (S81) is diagonal with x_4-diagonal entry c/σ_ε^2 (from $\mathbf{\Lambda}$). Hence

$$\kappa^2_{\text{Bayesian}}(\text{Study 3}, x_4) = \frac{1}{M} \cdot \frac{\sigma_\varepsilon^2}{c} = \frac{\sigma_\varepsilon^2}{Mc} = \frac{\sigma_\varepsilon^2}{N-1}, \tag{S82}$$

which is independent of the numerical value of $\tau^2_{0,\text{working}}$: no choice of working intercept variance can repair the omitted between-cluster contribution, because the correction term structurally cannot reach the x_4-coordinate.

Conclusion. Equations (S82) and (S73) show that, although Simulation Studies 2 and 3 share the same inference target for x_4, their Bayesian variance differs by exactly the between-cluster random-slope contribution τ_4^2/M. In Simulation Study 2, the working random-effects structure correctly includes the random slope of x_4, and the Bayesian variance therefore coincides with the true sampling variance of T_4. In Simulation Study 3, this random slope is omitted from the working model, so the corresponding between-cluster component is absent from the Bayesian variance, regardless of the working random-intercept variance. This explains why the Bayesian conditional procedure remains calibrated in Simulation Study 2 but produces liberal Type-I error rates for x_4 in Simulation Study 3.

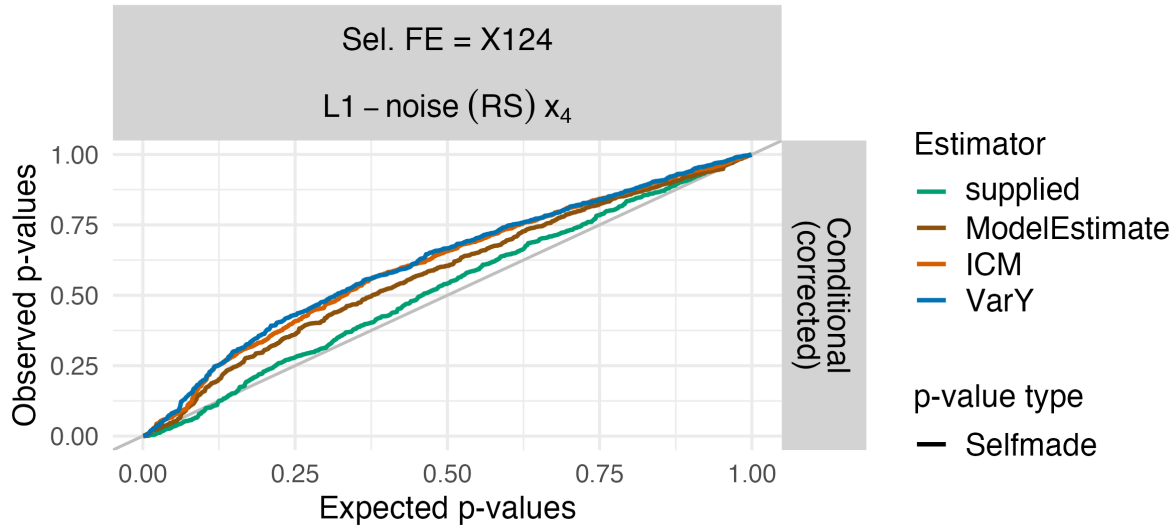
This difference, however, should not be interpreted as evidence that the conditional perspective is intrinsically invalid under random-effects misspecification. The conditional perspective is formulated for inference conditional on the realized random-effects vector $\mathbf{b}$, with a corresponding realization-specific inference target. A strictly conditional calibration analysis would therefore hold $\mathbf{b}$ fixed across Monte Carlo replications. In our simulations, by contrast, $\mathbf{b}$ is redrawn in every replication. The reported Type-I error rates consequently reflect variability across both the level-1 errors and the random-effects distribution. Under this repeated-sampling framework, the omitted x_4 random slope in Simulation Study 3 contributes to the variability of the test statistic but this contribution is not represented in the Bayesian variance of the misspecified working model. Thus, the observed failure concerns the application of the conditional procedure within our repeated-sampling framework under random-effects misspecification, rather than the validity of the conditional perspective within its intended conditioning framework.

Fixing the Random-Effects Vector in Simulation Study 3. To further support our explanation above that the conditional perspective is not generally invalid, we repeated one of the simulation conditions from Simulation Study 3 while holding the random-effects vector fixed across Monte Carlo replications. Specifically, we generated the random-effects vector $\mathbf{b}$ once and reused the same realization in all replications, so that only the level-1 residuals were newly sampled in each replication. We focused on inference for the fixed effect of x_4 under the selection outcome $\mathcal{A} = \{1, 2, 4\}$ and evaluated only the corrected Bayesian conditional perspective.

Figure S1 presents the corresponding Q–Q-plots for tests against the conditional target $\bar{b}_{4,M}$. Under the supplied covariance matrices $\mathbf{G}_{\text{working}}$ and $\mathbf{R}$, the resulting p-values were approximately uniformly distributed (slightly conservative). When the covariance structures were estimated, inference became more conservative.

Figure S1

Simulation Study 3: Conditional Inference with a Fixed Random-Effects Vector



Note. The figure shows Q–Q plots from a repetition of Simulation Study 3 under the corrected Bayesian conditional perspective with the random-effects vector held fixed across the 1,000 iterations. Columns display the observed p-values for the fixed effect of the Level-1 working-noise variable x_4 against standard uniform quantiles. Colors indicate the covariance estimators. Deviations toward the lower right (upper left) indicate liberal (conservative) p-values.

These results show that the corrected Bayesian conditional perspective provides approximately valid post-selection inference when testing against the realization-specific target

$\bar{b}_{4,M}$ in the considered setting. However, the additional simulation changes both the treatment of the random-effects vector and the null hypothesis being tested. In the original Simulation Study 3, the random-effects vector was redrawn across replications, and inference was evaluated against the marginal null hypothesis $\mathbb{E}_{\mathbf{b}}[\rho_4(\mathbf{b})] = 0$. Although this marginal null hypothesis holds, the corresponding conditional null hypothesis $\rho_4(\mathbf{b}) = 0$ generally does not hold for a given realization of $\mathbf{b}$. Thus, the liberal p-values observed in the original Simulation Study 3 should be interpreted in the context of testing the marginal null hypothesis with a conditional procedure under random-effects misspecification, rather than as evidence that the conditional perspective is intrinsically invalid.

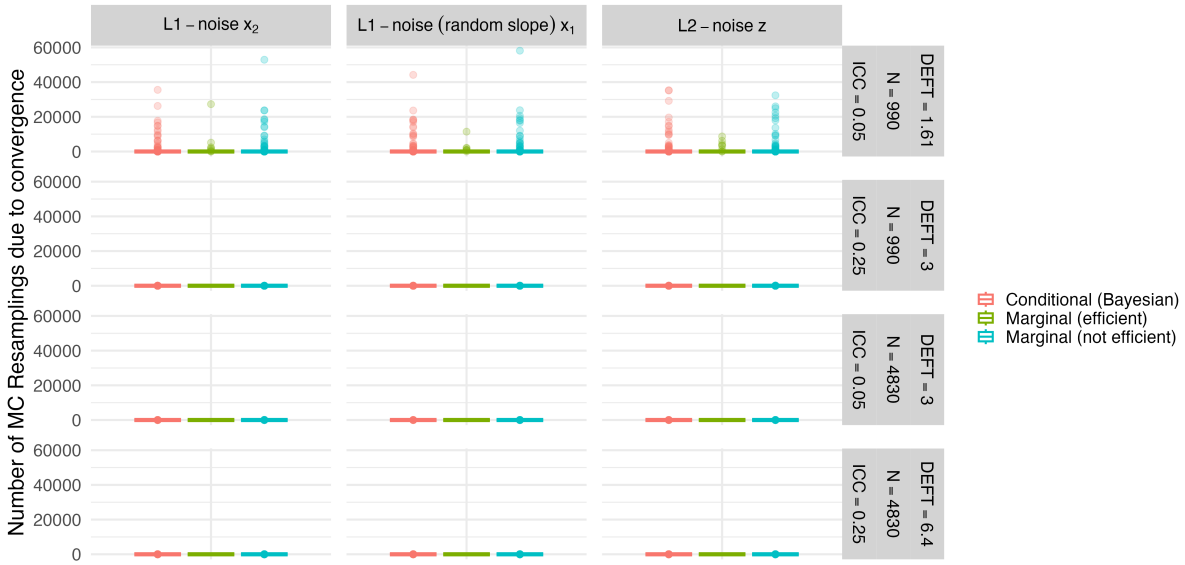
Convergence Issues in Simulation Studies 1–4

Convergence Issues in Simulation Study 1

In the condition with $N = 990$ and $ICC_{\text{pseudo}} = 0.05$, only 2,436 of the 3,000 initially simulated samples converged during model fitting. Each remaining sample had to be resampled at least once before convergence was achieved. In every other simulation condition, at most 2 of the 3,000 samples exhibited convergence issues and required resampling. A similar pattern emerged during MC sampling (see Figure S2, which shows the number of resamples required to obtain 1,000 samples without convergence issues).

Figure S2

Simulation Study 1: Number of MC Resamples Required Because of Convergence Issues



Note. The figure illustrates the extent of convergence issues during MC sampling for the selfmade approach in Simulation Study 1. The boxplots show the number of resamples required because of convergence issues, separately for the fixed effects of the noise predictors across the four simulation conditions. Colors indicate the inferential perspective used.

Convergence Issues in Simulation Study 2

In Simulation Study 2, the outcome variable was resampled until the full linear candidate model converged, the fixed-effects structure of interest was selected, and the selected model converged without issues. Convergence issues occurred in approximately 17 % of fits of the full linear candidate model. Conditional on convergence of the full model and selection of at least one noise predictor of interest, the selected model exhibited convergence issues in approximately 6 % of cases.

Figure S3 shows the number of MC resamples required to obtain 1,000 samples without convergence issues for each condition in which additional resampling occurred. The results are presented separately for each selected fixed-effects working model, predictor, inferential perspective, and covariance matrix estimator. They were very similar across covariance matrix estimators and inferential perspectives. Most simulation samples required a moderate number of resamples (75 % quantile: 134.00; 95 % quantile: 581.05), although a few extreme outliers occurred.

Figure S3

Simulation Study 2: Number of MC Resamples Required Because of Convergence Issues



Note. The figure illustrates the extent of convergence issues during MC sampling for the selfmade approach in Simulation Study 2. Columns show boxplots of the number of resamples required because of convergence issues for the Level-1 working noise variables (x_3 and x_4). Rows distinguish the inferential perspectives used in the selfmade approach, and colors indicate covariance estimators. For readability, an additional row shows the not-efficient marginal perspective with the proposed covariance matrix estimators.

Singular Fits in Simulation Study 2

Simulation runs for which inversion of the estimated random-effects covariance matrix failed were retried. The frequency of these failures was not recorded. The results of Simulation Study 2 were consistent with those of the other simulation studies and did not suggest that these failures affected the observed inferential patterns.

Convergence Issues in Simulation Study 3

Convergence issues in Simulation Study 3 were negligible: fewer than 0.001% of full candidate model fits failed. Conditional on convergence of the full model and selection of at least one noise predictor of interest, the selected model exhibited convergence issues in fewer than 0.001 % of cases, and at most one MC sample per 1,000 required resampling.

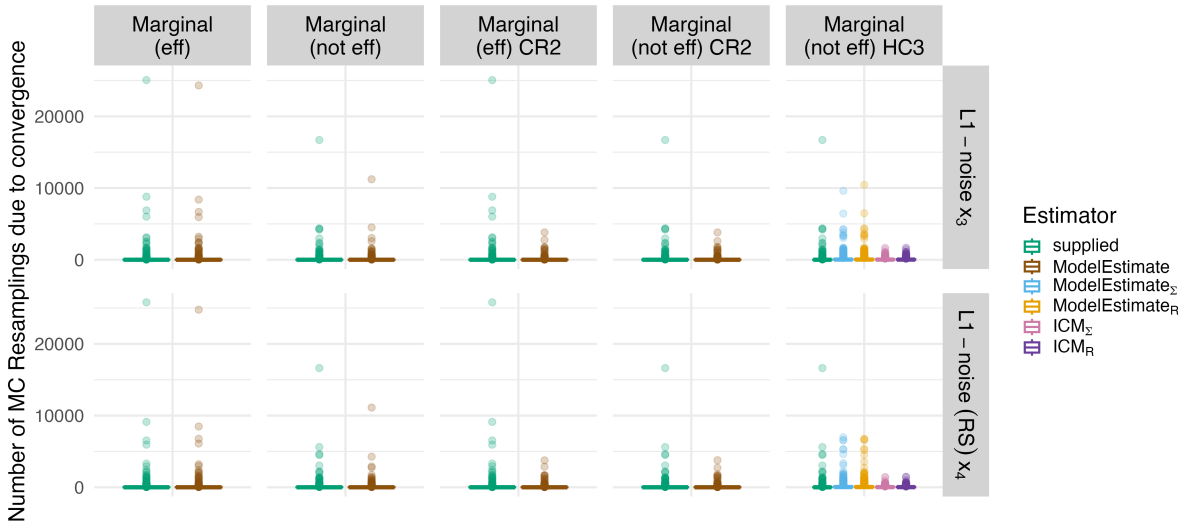
Convergence Issues in Simulation Study 4

In Simulation Study 4, the outcome variable was resampled until model (i) converged and model (ii) was selected and fitted without convergence issues. Convergence problems occurred in approximately 26 % of the initial fits of model (i) and in fewer than 0.001 % of the selected model fits, conditional on convergence of model (i).

Figure S4 shows the number of MC resamples required to obtain 1,000 samples without convergence issues, separately for each condition in which additional resampling occurred. Most cases required a moderate number of resamples.

Figure S4

Simulation Study 4: Number of MC Resamples Required Because of Convergence Issues



Note. The figure illustrates the extent of convergence issues during MC sampling for the selfmade approach in Simulation Study 4. Rows show boxplots of the number of resamples required because of convergence issues for the Level-1 working noise variables (x_3 and x_4). For readability additional columns show the extent of convergence issues during MC sampling for the selfmade approach under the robust approaches. Colors indicate the covariance matrix estimators used.

Iterations with no surviving Monte Carlo samples

In Simulation Study 4, we identified at most five iterations per condition in which no Monte Carlo samples survived the inference-region check for the HC3 covariance-matrix estimators. Consequently, p-values could not be computed for these cases. Ideally, a minimum number of surviving samples should have been specified in advance, and sampling should have continued until that threshold was reached. We did not implement such a criterion, consistent with the procedure used by Rügamer et al. (2022). Given the small number of affected iterations, these cases are unlikely to have materially affected the overall inferential patterns.

Potential Impact of Convergence and Singular Fits

Across the simulation studies, singular fits were common, and Simulation Study 2 exhibited a non-negligible number of convergence failures. To handle this, resampling was introduced until convergence was achieved. Excluding samples with convergence failures can, in principle, introduce bias if failures are related to data characteristics, selection outcomes, or test statistics. In Simulation Study 2, convergence failures occurred with comparable frequency across conditions and were not concentrated in settings where the Selfmade approach performed poorly. Although this does not rule out bias within individual conditions, we consider it unlikely that convergence failures substantially affected our results. This interpretation is supported by the negligible convergence problems in Simulation Study 3 and most conditions of Simulation Study 1, as well as the consistency of findings between Simulation Studies 2 and 3.[3]

[3] This is particularly evident in the repetition of Simulation Study 3 under the uncorrected Bayesian conditional perspective.

Singularity of Q and $\hat{\boldsymbol{\Sigma}}$

As noted in the main text, the newly proposed covariance matrix estimators can be applied directly within the not-efficient marginal framework. Their use within the efficient marginal framework, however, requires $\hat{\boldsymbol{\Sigma}}$ to be invertible because both the test vector and the variance of the efficient marginal test statistic depend on $\hat{\boldsymbol{\Sigma}}^{-1}$. We therefore examine in detail the conditions under which $\hat{\boldsymbol{\Sigma}}$ is singular in our simulation studies, which assume a balanced design, and summarize the alternative strategies we explored to address this problem.

For the HC0 and HC1 corrections, we show that $\boldsymbol{Q}_m$, and consequently $\boldsymbol{Q}$, is not invertible. The corresponding estimator $\hat{\boldsymbol{\Sigma}}$ is therefore singular, at least when $\hat{\boldsymbol{\Sigma}} = \boldsymbol{Q}$ or when the working random-effects structure contains only a random intercept:

Proposition 2. *If the HC0 or the HC1 correction is used, then* $\operatorname{rank}(\boldsymbol{Q}) \leq M$.

Proof. By construction, $\boldsymbol{Q} = \operatorname{diag}(\boldsymbol{Q}_1, \ldots, \boldsymbol{Q}_m, \ldots, \boldsymbol{Q}_M)$ is block-diagonal. Hence,

$$\operatorname{rank}(\boldsymbol{Q}) = \sum_{m=1}^{M} \operatorname{rank}(\boldsymbol{Q}_m). \tag{S83}$$

It therefore suffices to show that $\operatorname{rank}(\boldsymbol{Q}_m) \leq 1$ for each $m \in \{1, \ldots, M\}$. Without loss of generality, we first consider the HC0 correction. Under the HC1 correction, $\boldsymbol{Q}_m$ is additionally premultiplied by a scalar matrix. Because the rank of a matrix product is bounded above by the minimum rank of its factors, this additional multiplication does not affect the rank argument.

An arbitrary column c of $\boldsymbol{Q}_m$ has the form

$$\left(\hat{\varepsilon}_{1m}^{LM} \cdot \hat{\varepsilon}_{cm}^{LM}, \hat{\varepsilon}_{2m}^{LM} \cdot \hat{\varepsilon}_{cm}^{LM}, \ldots, \hat{\varepsilon}_{N_m m}^{LM} \cdot \hat{\varepsilon}_{cm}^{LM}\right)^{\top}. \tag{S84}$$

This column can be expressed as a scalar multiple of the first column,

$$\left(\hat{\varepsilon}_{1m}^{LM} \cdot \hat{\varepsilon}_{1m}^{LM}, \hat{\varepsilon}_{2m}^{LM} \cdot \hat{\varepsilon}_{1m}^{LM}, \ldots, \hat{\varepsilon}_{N_m m}^{LM} \cdot \hat{\varepsilon}_{1m}^{LM}\right)^{\top}, \tag{S85}$$

with scaling factor

$$\lambda_c = \frac{\hat{\varepsilon}_{cm}^{LM}}{\hat{\varepsilon}_{1m}^{LM}}, \tag{S86}$$

provided that $\hat{\varepsilon}_{1m}^{LM} \neq 0$. If $\hat{\varepsilon}_{1m}^{LM} = 0$, we choose any column index j for which $\hat{\varepsilon}_{jm}^{LM} \neq 0$; such an index exists whenever $\boldsymbol{Q}_m \neq \mathbf{0}$. Repeating the argument for this column shows that all columns of $\boldsymbol{Q}_m$ are linearly dependent and hence that $\text{rank}(\boldsymbol{Q}_m) \leq 1$. □

Under either the HC0 or HC1 correction, $\text{rank}(\boldsymbol{Q})$ is smaller than the number of observations, provided that $N > M$. Thus, $\boldsymbol{Q}$ is not invertible, and neither is $\hat{\boldsymbol{\Sigma}}$ when $\boldsymbol{Q}$ is used as its estimator. Moreover, if $\boldsymbol{Q}$ is used as $\hat{\boldsymbol{R}}$ and the random-effects structure contains only a random intercept, the resulting covariance matrix estimator $\hat{\boldsymbol{\Sigma}}$ is not invertible:

Corollary 1. *If the random-effects structure contains only a random intercept and the HC0 or the HC1 correction is used, then the matrix* $\hat{\boldsymbol{\Sigma}} = \boldsymbol{Z}_{\mathcal{B}}\hat{\boldsymbol{G}}\boldsymbol{Z}_{\mathcal{B}}^{\top} + \boldsymbol{Q}$ *is not invertible if* $N > 2M$.

Proof. By proposition 2, we have

$$\text{rank}(\boldsymbol{Q}) \leq M. \tag{S87}$$

If the random-effects structure contains only a random intercept, then $\boldsymbol{Z}_{\mathcal{B}}\hat{\boldsymbol{G}}\boldsymbol{Z}_{\mathcal{B}}^{\top}$ is block-diagonal, with one block per Level-2 unit. Each block has rank at most one. Consequently,

$$\text{rank}(\boldsymbol{Z}_{\mathcal{B}}\hat{\boldsymbol{G}}\boldsymbol{Z}_{\mathcal{B}}^{\top}) \leq M. \tag{S88}$$

By subadditivity of matrix rank,

$$\text{rank}(\hat{\boldsymbol{\Sigma}}) = \text{rank}\big(\boldsymbol{Z}_{\mathcal{B}}\hat{\boldsymbol{G}}\boldsymbol{Z}_{\mathcal{B}}^{\top} + \boldsymbol{Q}\big) \leq \text{rank}(\boldsymbol{Z}_{\mathcal{B}}\hat{\boldsymbol{G}}\boldsymbol{Z}_{\mathcal{B}}^{\top}) + \text{rank}(\boldsymbol{Q}) \leq 2M. \tag{S89}$$

Since $N > 2M$, it follows that

$$\text{rank}(\hat{\boldsymbol{\Sigma}}) < N. \tag{S90}$$

Hence, $\hat{\boldsymbol{\Sigma}}$ is not invertible. □

In Simulation Study 3, more than two Level-1 units were nested within each Level-2 unit; hence, $N > 2M$. Therefore, $\hat{\boldsymbol{\Sigma}}$ was not invertible in Simulation Study 3.

Because no general rank equality or inequality is available for Hadamard (element-wise) division, we cannot derive a general statement about the singularity of $\boldsymbol{Q}$ and, consequently, of $\hat{\boldsymbol{\Sigma}}$ under the HC2 or HC3 correction without knowledge about the predictor

matrix $\mathbf{X}$. For a given selected predictor matrix, however, we can establish the following sufficient condition for the singularity of $\hat{\boldsymbol{\Sigma}}$.

Corollary 2. *Suppose that an HCk correction with $k \in \{2, 3\}$ is used and that*

$$\operatorname{rank}\left(\mathbf{1} \oslash (\mathbf{1} - (\mathbf{X}_{\mathcal{A}}(\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A}}^{\top}))^{\circ k-1}\right) < \frac{N}{M}, \tag{S91}$$

then $\hat{\boldsymbol{\Sigma}}$ is not invertible if $\hat{\boldsymbol{\Sigma}} = \boldsymbol{Q}$.[4] *Furthermore, if*

$$\operatorname{rank}\left(\mathbf{1} \oslash (\mathbf{1} - (\mathbf{X}_{\mathcal{A}}(\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A}}^{\top}))^{\circ k-1}\right) < \frac{N - M}{M} \tag{S92}$$

then $\hat{\boldsymbol{\Sigma}}$ is not invertible if $\hat{\boldsymbol{R}} = \boldsymbol{Q}$ and the random-effects structure contains only a random intercept.

Proof. For an HCk correction with $k \in \{2, 3\}$, we have

$$\begin{aligned}
\operatorname{rank}(\boldsymbol{Q}) &\leq \operatorname{rank}\left(\boldsymbol{Q}_{\boldsymbol{HC0}} \oslash (\mathbf{1} - (\mathbf{X}_{\mathcal{A}}(\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A}}^{\top}))^{\circ k-1}\right) \\
&\overset{\text{rank inequality for the Hadamard product}}{\leq} \operatorname{rank}\left(\mathbf{1} \oslash (\mathbf{1} - (\mathbf{X}_{\mathcal{A}}(\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A}}^{\top}))^{\circ k-1}\right) \cdot \operatorname{rank}(\boldsymbol{Q}_{\boldsymbol{HC0}}) \\
&\overset{\text{proposition 2}}{\leq} \operatorname{rank}\left(\mathbf{1} \oslash (\mathbf{1} - (\mathbf{X}_{\mathcal{A}}(\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A}}^{\top}))^{\circ k-1}\right) \cdot M.
\end{aligned}$$

Thus, if

$$\operatorname{rank}\left(\mathbf{1} \oslash (\mathbf{1} - (\mathbf{X}_{\mathcal{A}}(\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A}}^{\top}))^{\circ k-1}\right) < \frac{N}{M}, \tag{S93}$$

then $\operatorname{rank}(\boldsymbol{Q}) < N$. Consequently, $\boldsymbol{Q}$ does not have full rank and is not invertible.

Similarly, if

$$\operatorname{rank}\left(\mathbf{1} \oslash (\mathbf{1} - (\mathbf{X}_{\mathcal{A}}(\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A}}^{\top}))^{\circ k-1}\right) < \frac{N - M}{M}, \tag{S94}$$

[4] $\oslash$ denotes the Hadamard division and $^{\circ(k-1)}$ the $(k-1)$-st Hadamard power.

then, by subadditivity of matrix rank,

$$\begin{aligned} \operatorname{rank}\left(\boldsymbol{Z}_{\mathcal{B}}\hat{\boldsymbol{G}}\boldsymbol{Z}_{\mathcal{B}}^{\top} + \boldsymbol{Q}\right) &\leq \operatorname{rank}(\boldsymbol{Z}_{\mathcal{B}}\hat{\boldsymbol{G}}\boldsymbol{Z}_{\mathcal{B}}^{\top}) + \operatorname{rank}(\boldsymbol{Q}) \\ &\overset{\operatorname{rank}(\boldsymbol{Z}_{\mathcal{B}}\hat{\boldsymbol{G}}\boldsymbol{Z}_{\mathcal{B}}^{\top}) \leq M.}{<} M + \frac{N-M}{M} \cdot M = N. \end{aligned} \tag{S95}$$

□

Simulation Studies 2, 3, and 4 each used a single fixed predictor matrix $\mathbf{X}$. We could therefore compute the numerical rank of the corresponding Hadamard-division matrix, based on the hat matrix, for every possible selected predictor matrix $\mathbf{X}_{\mathcal{A}}$ (i.e., every submatrix of the fixed $\mathbf{X}$). [5] The fixed predictor matrices used in Simulation Studies 2, 3, and 4 are provided in our GitHub repository, allowing the reported rank calculations to be reproduced. These calculations indicated that the HC2-based estimator $\hat{\boldsymbol{\Sigma}}$ was numerically rank-deficient for every selected predictor matrix considered in Simulation Study 3, irrespective of whether $\hat{\boldsymbol{\Sigma}} = \boldsymbol{Q}$ or $\hat{\boldsymbol{R}} = \boldsymbol{Q}$.

For the HC3 correction, the sufficient condition for singularity established above implies that $\hat{\boldsymbol{\Sigma}}$ is rank-deficient whenever $\hat{\boldsymbol{\Sigma}} = \boldsymbol{Q}$ for any predictor matrix that may be selected in Simulation Study 3. When $\hat{\boldsymbol{R}} = \boldsymbol{Q}$, this sufficient condition and the computed numerical ranks of the corresponding Hadamard-division matrices indicate rank deficiency whenever the selected model contains fewer predictors than $\mathbf{X}$. If the full set of predictors is selected, however, the rank condition alone does not establish singularity. We therefore examined a set of random samples in which the full linear model was selected and computed the numerical rank of $\hat{\boldsymbol{\Sigma}} = \mathbf{Z}_{\mathcal{B}}\hat{\mathbf{G}}\mathbf{Z}_{\mathcal{B}}^{\top} + \boldsymbol{Q}_{\mathrm{HC3}}$. The matrix was numerically rank-deficient in the examined samples. These results suggest that $\hat{\boldsymbol{\Sigma}}$ may also be numerically rank-deficient when the full model is selected; however, they do not establish that this holds for every possible sample.

We considered the pseudoinverse and Tikhonov regularization of $\hat{\boldsymbol{\Sigma}}$ as potential strategies for making the efficient marginal perspective feasible with our estimators. However, pilot simulations indicated that both approaches still yielded liberal p-values. We therefore restrict the proposed covariance matrix estimators to the not-efficient marginal framework.

[5] Because of numerical issues, these calculations were performed in MATLAB (Inc., 2025) rather than in R (R Core Team, 2025).

Construction of the CR-Based Variance Estimators

This supplement details the construction of the cluster-robust (CR0 and CR2) variance estimators introduced in Simulation Study 3 to replace κ^2 and κ^2_{eff}.

Not-Efficient Marginal Perspective

Let $\hat{\boldsymbol{\beta}}_{\mathcal{A}} = (\mathbf{X}_{\mathcal{A}}^\top \mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A}}^\top \mathbf{y}_{\text{obs}}$ denote the OLS estimator from the auxiliary LM, and $\hat{\boldsymbol{\varepsilon}}^{\text{LM}} = \mathbf{y}_{\text{obs}} - \mathbf{X}_{\mathcal{A}}\hat{\boldsymbol{\beta}}_{\mathcal{A}}$ the corresponding residual vector. For cluster m, let $\mathbf{X}_{\mathcal{A},m}$ denote the rows of $\mathbf{X}_{\mathcal{A}}$ and $\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}}$ the residuals belonging to cluster m, and let

$$\mathbf{H}_m = \mathbf{X}_{\mathcal{A},m}(\mathbf{X}_{\mathcal{A}}^\top \mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A},m}^\top \tag{S96}$$

denote the corresponding diagonal block of the OLS hat matrix.

The cluster-robust sandwich (*CR0*) estimator (Liang & Zeger, 1986) uses the raw residuals directly,

$$\widehat{\mathbf{K}}_{\text{CR0}}(\hat{\boldsymbol{\beta}}_{\mathcal{A}}) = (\mathbf{X}_{\mathcal{A}}^\top \mathbf{X}_{\mathcal{A}})^{-1}\left(\sum_{m=1}^{M} \mathbf{X}_{\mathcal{A},m}^\top \hat{\boldsymbol{\varepsilon}}_m^{\text{LM}}(\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}})^\top \mathbf{X}_{\mathcal{A},m}\right)(\mathbf{X}_{\mathcal{A}}^\top \mathbf{X}_{\mathcal{A}})^{-1}. \tag{S97}$$

Because raw OLS residuals are, on average, attenuated relative to the true errors, CR0 tends to underestimate the true sampling variability, particularly with few clusters (Huang et al., 2023; Maas & Hox, 2005).

The *CR2* estimator of Bell and McCaffrey (2002) corrects for this by adjusting the cluster-specific residuals with a matrix that depends on the corresponding block of the hat matrix, prior to forming the sandwich estimator. The adjustment matrix is the symmetric inverse square root of $\mathbf{I}_{N_m} - \mathbf{H}_m$,

$$\mathbf{Adj}_m = \left[\mathbf{I}_{N_m} - \mathbf{H}_m\right]^{-1/2}. \tag{S98}$$

The resulting CR2 adjustment of the cluster-specific residual cross-products is

$$\widehat{\boldsymbol{\Omega}}_m = \mathbf{Adj}_m\, \hat{\boldsymbol{\varepsilon}}_m^{\text{LM}}(\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}})^\top \mathbf{Adj}_m. \tag{S99}$$

The CR2 sandwich estimator is then obtained analogously to (S97), replacing $\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}}(\hat{\boldsymbol{\varepsilon}}_m^{\text{LM}})^\top$

with $\widehat{\boldsymbol{\Omega}}_m$:

$$\widehat{\mathbf{K}}_{\mathrm{CR2}}(\hat{\boldsymbol{\beta}}_{\mathcal{A}}) = (\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}\left(\sum_{m=1}^{M}\mathbf{X}_{\mathcal{A},m}^{\top}\,\widehat{\boldsymbol{\Omega}}_m\,\mathbf{X}_{\mathcal{A},m}\right)(\mathbf{X}_{\mathcal{A}}^{\top}\mathbf{X}_{\mathcal{A}})^{-1}. \tag{S100}$$

This unweighted construction matches the OLS-based structure of T and its null variance κ^2 in Equation (17), and does not involve $\boldsymbol{\Sigma}$. We substitute

$$\hat{\kappa}^2_{\mathrm{CR0}} = \mathbf{e}_j^{\top}\,\widehat{\mathbf{K}}_{\mathrm{CR0}}(\hat{\boldsymbol{\beta}}_{\mathcal{A}})\,\mathbf{e}_j, \qquad \hat{\kappa}^2_{\mathrm{CR2}} = \mathbf{e}_j^{\top}\,\widehat{\mathbf{K}}_{\mathrm{CR2}}(\hat{\boldsymbol{\beta}}_{\mathcal{A}})\,\mathbf{e}_j \tag{S101}$$

for κ^2 in Equation (17); in the main text we report results based on $\hat{\kappa}^2_{\mathrm{CR2}}$.

Efficient Marginal Perspective

The efficient marginal statistic T_{eff} is based on the $\boldsymbol{\Sigma}^{-1}$-weighted GLS estimator, with null variance κ^2_{eff} given in Equation (19). A robust variance estimator matching this weighting must therefore itself incorporate $\boldsymbol{\Sigma}^{-1}$, rather than relying on the unweighted hat matrix used above. Following the extension of the CR estimator to mixed-model settings (Cheong et al., 2001; Huang et al., 2023; McCaffrey et al., 2001; Pustejovsky, 2026; Pustejovsky & Tipton, 2018; Tipton, 2015), define the weighted hat matrix as:

$$\mathbf{H}_m^{\mathrm{eff}} = \mathbf{X}_{\mathcal{A},m}\left(\mathbf{X}_{\mathcal{A}}^{\top}\boldsymbol{\Sigma}^{-1}\mathbf{X}_{\mathcal{A}}\right)^{-1}\mathbf{X}_{\mathcal{A},m}^{\top}\boldsymbol{\Sigma}_m^{-1}. \tag{S102}$$

Let $\hat{\boldsymbol{\beta}}_{\mathcal{A},\mathrm{eff}} = (\mathbf{X}_{\mathcal{A}}^{\top}\boldsymbol{\Sigma}^{-1}\mathbf{X}_{\mathcal{A}})^{-1}\mathbf{X}_{\mathcal{A}}^{\top}\boldsymbol{\Sigma}^{-1}\mathbf{y}_{\mathrm{obs}}$ denote the efficient/GLS estimator from the auxiliary LM, and let

$$\hat{\boldsymbol{\varepsilon}}^{\mathrm{LM}}_{\mathrm{eff},m} = \mathbf{y}_{\mathrm{obs},m} - \mathbf{X}_{\mathcal{A},m}\hat{\boldsymbol{\beta}}_{\mathcal{A},\mathrm{eff}} \tag{S103}$$

denote the corresponding GLS residuals for cluster m. Analogous to (S97), the unadjusted (*CR0*) estimator for the efficient perspective is

$$\begin{aligned}&\widehat{\mathbf{K}}^{\mathrm{eff}}_{\mathrm{CR0}}(\hat{\boldsymbol{\beta}}_{\mathcal{A},\mathrm{eff}})\\ &= \left(\mathbf{X}_{\mathcal{A}}^{\top}\boldsymbol{\Sigma}^{-1}\mathbf{X}_{\mathcal{A}}\right)^{-1}\left(\sum_{m=1}^{M}\mathbf{X}_{\mathcal{A},m}^{\top}\boldsymbol{\Sigma}_m^{-1}\,\hat{\boldsymbol{\varepsilon}}^{\mathrm{LM}}_{\mathrm{eff},m}\left(\hat{\boldsymbol{\varepsilon}}^{\mathrm{LM}}_{\mathrm{eff},m}\right)^{\top}\boldsymbol{\Sigma}_m^{-1}\mathbf{X}_{\mathcal{A},m}\right)\left(\mathbf{X}_{\mathcal{A}}^{\top}\boldsymbol{\Sigma}^{-1}\mathbf{X}_{\mathcal{A}}\right)^{-1}.\end{aligned} \tag{S104}$$

For the bias-reduced (*CR2*) adjustment in this weighted setting, we use the more general adjustment matrix proposed by McCaffrey et al. (2001) (see also footnote 5 in Huang et al.

(2023)), which additionally incorporates $\mathbf{\Sigma}_m$, the cluster-m diagonal block of $\mathbf{\Sigma}$, via its Cholesky root. Specifically, let $\mathbf{U}_m$ denote the upper-triangular matrix satisfying $\mathbf{\Sigma}_m = \mathbf{U}_m^\top \mathbf{U}_m$, and define

$$\mathbf{B}_m^{\text{eff}} = \mathbf{U}_m \left[\mathbf{I}_{N_m} - \mathbf{H}_m^{\text{eff}}\right] \mathbf{\Sigma}_m \, \mathbf{U}_m^\top, \qquad \mathbf{Adj}_m^{\text{eff}} = \mathbf{U}_m^\top \left(\mathbf{B}_m^{\text{eff}}\right)^{-\frac{1}{2}} \mathbf{U}_m. \tag{S105}$$

The adjusted cross-products are

$$\widehat{\mathbf{\Omega}}_m^{\text{eff}} = \mathbf{Adj}_m^{\text{eff}} \, \hat{\boldsymbol{\varepsilon}}_{\text{eff},m}^{\text{LM}} \left(\hat{\boldsymbol{\varepsilon}}_{\text{eff},m}^{\text{LM}}\right)^\top \mathbf{Adj}_m^{\text{eff}}, \tag{S106}$$

and the corresponding CR2 sandwich estimator is

$$\widehat{\mathbf{K}}_{\text{CR2}}^{\text{eff}}(\hat{\boldsymbol{\beta}}_{\mathcal{A},\text{eff}}) = \left(\mathbf{X}_{\mathcal{A}}^\top \mathbf{\Sigma}^{-1} \mathbf{X}_{\mathcal{A}}\right)^{-1} \left(\sum_{m=1}^{M} \mathbf{X}_{\mathcal{A},m}^\top \mathbf{\Sigma}_m^{-1} \, \widehat{\mathbf{\Omega}}_m^{\text{eff}} \, \mathbf{\Sigma}_m^{-1} \mathbf{X}_{\mathcal{A},m}\right) \left(\mathbf{X}_{\mathcal{A}}^\top \mathbf{\Sigma}^{-1} \mathbf{X}_{\mathcal{A}}\right)^{-1}. \tag{S107}$$

Here, $\mathbf{\Sigma}$ enters only as a working matrix within the weighting and adjustment of the sandwich construction, in exactly the same role it already plays in defining $\mathbf{v}_{\text{eff}}$ and κ_{eff}^2; it is not altered or replaced, and the selfmade transformation itself, which depends on $\mathbf{\Sigma}$ through $\mathbf{v}_{\text{eff}}$ and $\mathbf{dir}_{\boldsymbol{v}}$, remains untouched. We substitute

$$\hat{\kappa}_{\text{eff,CR0}}^2 = \mathbf{e}_j^\top \, \widehat{\mathbf{K}}_{\text{CR0}}^{\text{eff}}(\hat{\boldsymbol{\beta}}_{\mathcal{A},\text{eff}}) \, \mathbf{e}_j, \qquad \hat{\kappa}_{\text{eff,CR2}}^2 = \mathbf{e}_j^\top \, \widehat{\mathbf{K}}_{\text{CR2}}^{\text{eff}}(\hat{\boldsymbol{\beta}}_{\mathcal{A},\text{eff}}) \, \mathbf{e}_j \tag{S108}$$

for κ_{eff}^2 in Equation (19); as in the not-efficient case, the main text reports results based on the bias-reduced estimate $\hat{\kappa}_{\text{eff,CR2}}^2$.

Choice of the Covariance Matrix Estimator

The not-efficient CR0 and CR2 estimators do not involve $\mathbf{\Sigma}$ themselves; wherever $\hat{\mathbf{\Sigma}}$ is still required in the not-efficient perspective, we always use the *ModelEstimate* estimator of Rügamer et al. (2022). In the efficient perspective, $\mathbf{\Sigma}$ is generally unknown in practice and must be estimated. Since $\mathbf{\Sigma}$ enters the CR construction only as a working matrix rather than as the object being replaced, any of the standard covariance estimators (*ModelEstimate*, *ICM*) for $\mathbf{\Sigma}$ proposed by Rügamer et al. (2022) can equally be substituted into Equations (S102), (S105), and (S104)–(S107).

Equivalence of Bayesian and frequentist variance estimates for the fixed slopes in Simulation Study 3

To demonstrate that the failure of the corrected Bayesian conditional perspective in Simulation Study 3 under our simulation framework also occurs with the uncorrected Bayesian and frequentist conditional perspectives, we (1) show that the Bayesian and frequentist variance estimates for the fixed slopes coincide and (2) show that repeating Simulation Study 3 using the uncorrected Bayesian conditional perspective yields the same results as using the corrected Bayesian conditional perspective.

Proposition 3. *Consider a random-intercept LMM as in Simulation Study 3 with M Level-2 units and N_m observations in Level-2 unit m, such that $N = \sum_{m=1}^{M} N_m$. Suppose that the fixed-effects design matrix $\mathbf{X}_{\mathcal{A}} = (\mathbf{1}_N, \mathbf{X}_{\mathcal{A},1}, \ldots, \mathbf{X}_{\mathcal{A},P})$ contains a fixed intercept and exactly P Level-1 predictors, with no Level-2 predictors, and has full column rank $P + 1$. Moreover, suppose that*

$$\mathbf{G}_{working} = \tau_0^2 \mathbf{I}_M, \qquad \mathbf{R} = \sigma_\varepsilon^2 \mathbf{I}_N, \tag{S109}$$

where $\tau_0^2 > 0$ and $\sigma_\varepsilon^2 > 0$.

We formulate the argument in terms of the population (working) covariance matrices. The same reasoning applies to covariance estimates that preserve the assumed covariance structure and yield positive variance estimates. Let

$$\mathbf{Z}_{\mathcal{B}} = \operatorname{diag}\left(\mathbf{1}_{N_1}, \ldots, \mathbf{1}_{N_M}\right) \tag{S110}$$

denote the (working) random-effects design matrix.

Suppose further that, for each Level-1 predictor $\mathbf{X}_{\mathcal{A},j}$, its mean is equal to 2 in every Level-2 unit, such that

$$\mathbf{X}_{\mathcal{A},j} = 2\mathbf{1}_N + \widetilde{\mathbf{X}}_{\mathcal{A},j}, \tag{S111}$$

where $\widetilde{\mathbf{X}}_{\mathcal{A},mj}$ denotes the subvector corresponding to Level-2 unit m and

$$\mathbf{1}_{N_m}^\top \widetilde{\mathbf{X}}_{\mathcal{A},mj} = 0 \qquad \text{for all } m = 1, \ldots, M, \quad j = 1, \ldots, P. \tag{S112}$$

Then the Bayesian and frequentist variance quantities coincide for every Level-1 fixed-effect slope.

Proof. For clarity, we explicitly include the index j in the following expressions to emphasize that the argument applies to each fixed-effect slope separately. Let $\mathbf{e}_{j+1}^{(P+1+M)}$ denote the unit vector corresponding to the jth slope in the combined fixed- and random-effects parameter vector, where the intercept occupies the first coordinate. The Bayesian and frequentist variance quantities can be written as

$$\kappa^2_{\text{Bayesian},j} = \mathbf{e}_{j+1}^{(P+1+M)\top}\mathbf{K}^{-1}\mathbf{e}_{j+1}^{(P+1+M)} \tag{S113}$$

and

$$\kappa^2_{\text{freq},j} = \mathbf{e}_{j+1}^{(P+1+M)\top}\mathbf{K}^{-1}\mathbf{H}\mathbf{K}^{-1}\mathbf{e}_{j+1}^{(P+1+M)}, \tag{S114}$$

where

$$\mathbf{H} = \mathbf{C}^\top\mathbf{R}^{-1}\mathbf{C} = \begin{pmatrix} \mathbf{X}_{\mathcal{A}}^\top\mathbf{R}^{-1}\mathbf{X}_{\mathcal{A}} & \mathbf{X}_{\mathcal{A}}^\top\mathbf{R}^{-1}\mathbf{Z}_{\mathcal{B}} \\ \mathbf{Z}_{\mathcal{B}}^\top\mathbf{R}^{-1}\mathbf{X}_{\mathcal{A}} & \mathbf{Z}_{\mathcal{B}}^\top\mathbf{R}^{-1}\mathbf{Z}_{\mathcal{B}} \end{pmatrix}, \qquad \mathbf{C} = (\mathbf{X}_{\mathcal{A}}, \mathbf{Z}_{\mathcal{B}}), \tag{S115}$$

and

$$\mathbf{K} = \mathbf{H} + \mathbf{A}, \qquad \mathbf{A} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{G}_{\text{working}}^{-1} \end{pmatrix}. \tag{S116}$$

Since $\mathbf{H} = \mathbf{K} - \mathbf{A}$,

$$\mathbf{K}^{-1}\mathbf{H}\mathbf{K}^{-1} = \mathbf{K}^{-1} - \mathbf{K}^{-1}\mathbf{A}\mathbf{K}^{-1}. \tag{S117}$$

Consequently,

$$\kappa^2_{\text{Bayesian},j} - \kappa^2_{\text{freq},j} = \mathbf{e}_{j+1}^{(P+1+M)\top}\mathbf{K}^{-1}\mathbf{A}\mathbf{K}^{-1}\mathbf{e}_{j+1}^{(P+1+M)}. \tag{S118}$$

To see when this difference vanishes, write

$$\mathbf{k} := \mathbf{K}^{-1}\mathbf{e}_{j+1}^{(P+1+M)} = \begin{pmatrix} \mathbf{k}_{\mathcal{A}} \\ \mathbf{k}_{\mathcal{B}} \end{pmatrix}, \tag{S119}$$

where $\mathbf{k}_{\mathcal{A}}$ and $\mathbf{k}_{\mathcal{B}}$ are the fixed-effects and random-effects blocks, respectively. Then

$$\kappa^2_{\text{Bayesian},j} - \kappa^2_{\text{freq},j} = \mathbf{k}^\top \mathbf{A}\mathbf{k} = \mathbf{k}_{\mathcal{B}}^\top \mathbf{G}^{-1}_{\text{working}} \mathbf{k}_{\mathcal{B}}. \tag{S120}$$

Because $\mathbf{G}^{-1}_{\text{working}}$ is positive definite, the two variance expressions coincide if and only if

$$\mathbf{k}_{\mathcal{B}} = \mathbf{0}. \tag{S121}$$

To establish this for the fixed-effect slopes, consider the decomposition $\mathbf{X}_{\mathcal{A},j} = 2\mathbf{1}_N + \widetilde{\mathbf{X}}_{\mathcal{A},j}$. By the assumed cluster means and the definition of $\mathbf{R}$,

$$\mathbf{Z}_{\mathcal{B}}^\top \mathbf{R}^{-1} \widetilde{\mathbf{X}}_{\mathcal{A},j} = \sigma_\varepsilon^{-2} \begin{pmatrix} \mathbf{1}_{N_1}^\top \widetilde{\mathbf{X}}_{\mathcal{A},1j} \\ \vdots \\ \mathbf{1}_{N_M}^\top \widetilde{\mathbf{X}}_{\mathcal{A},Mj} \end{pmatrix} = \mathbf{0}. \tag{S122}$$

Thus, the centered components of the Level-1 predictors are orthogonal to the columns of $\mathbf{Z}_{\mathcal{B}}$ with respect to $\mathbf{R}^{-1}$.

The fact that $\mathbf{X}_{\mathcal{A},j}$ itself is not orthogonal to $\mathbf{Z}_{\mathcal{B}}$ does not contradict this result. Indeed,

$$\begin{aligned} \mathbf{Z}_{\mathcal{B}}^\top \mathbf{R}^{-1} \mathbf{X}_{\mathcal{A},j} &= 2\mathbf{Z}_{\mathcal{B}}^\top \mathbf{R}^{-1} \mathbf{1}_N + \mathbf{Z}_{\mathcal{B}}^\top \mathbf{R}^{-1} \widetilde{\mathbf{X}}_{\mathcal{A},j} \\ &= 2\mathbf{Z}_{\mathcal{B}}^\top \mathbf{R}^{-1} \mathbf{1}_N \neq \mathbf{0}. \end{aligned} \tag{S123}$$

However, the constant component $2\mathbf{1}_N$ can be absorbed into the intercept, since

$$\beta_0 \mathbf{1}_N + \sum_{j=1}^{P} \beta_j \mathbf{X}_{\mathcal{A},j} = \left(\beta_0 + 2 \sum_{j=1}^{P} \beta_j \right) \mathbf{1}_N + \sum_{j=1}^{P} \beta_j \widetilde{\mathbf{X}}_{\mathcal{A},j}. \tag{S124}$$

This reparameterization leaves all slope coefficients unchanged. Consequently, the variance quantities for the slopes can be obtained using the reparameterized fixed-effects design matrix

$$\widetilde{\mathbf{X}}_{\mathcal{A}} = \left(\mathbf{1}_N, \widetilde{\mathbf{X}}_{\mathcal{A},1}, \ldots, \widetilde{\mathbf{X}}_{\mathcal{A},P} \right). \tag{S125}$$

Formally, $\mathbf{X}_{\mathcal{A}} = \widetilde{\mathbf{X}}_{\mathcal{A}}\mathbf{T}$ with the invertible matrix

$$\mathbf{T} = \begin{pmatrix} 1 & 2 \cdot \mathbf{1}_P^\top \\ \mathbf{0} & \mathbf{I}_P \end{pmatrix}, \tag{S126}$$

so $\widetilde{\mathbf{X}}_{\mathcal{A}}$ has full column rank as well. With $\mathbf{D} = \mathrm{diag}(\mathbf{T}, \mathbf{I}_M)$ we get $\mathbf{H} = \mathbf{D}^\top \widetilde{\mathbf{H}}\mathbf{D}$ and $\mathbf{K} = \mathbf{D}^\top \widetilde{\mathbf{K}}\mathbf{D}$, where $\widetilde{\mathbf{H}}, \widetilde{\mathbf{K}}$ are built from $\widetilde{\mathbf{X}}_{\mathcal{A}}$ (the penalty $\mathbf{A}$ acts only on the random effects and is unchanged). Since $(\mathbf{D}^{-1})^\top \mathbf{e}_{j+1}^{(P+1+M)} = \mathbf{e}_{j+1}^{(P+1+M)}$, both $\kappa^2_{\mathrm{Bayesian},j}$ and $\kappa^2_{\mathrm{freq},j}$ are unchanged when $(\mathbf{H}, \mathbf{K})$ is replaced by $(\widetilde{\mathbf{H}}, \widetilde{\mathbf{K}})$. In the following, $\mathbf{H}$ and $\mathbf{K}$ denote $\widetilde{\mathbf{H}}$ and $\widetilde{\mathbf{K}}$. Because $\widetilde{\mathbf{X}}_{\mathcal{A}}$ has full column rank and $\mathbf{G}^{-1}_{\mathrm{working}}$ is positive definite, $\mathbf{K}$ is positive definite, so $\mathbf{K}_{22}$ and the Schur complement $\mathbf{S}$ are invertible.

By construction,

$$\mathbf{Z}_{\mathcal{B}}^\top \mathbf{R}^{-1} \left(\widetilde{\mathbf{X}}_{\mathcal{A},1} \quad \cdots \quad \widetilde{\mathbf{X}}_{\mathcal{A},P} \right) = \mathbf{0}. \tag{S127}$$

To see why this orthogonality forces $\mathbf{k}_{\mathcal{B}} = \mathbf{0}$, partition $\mathbf{K}$ as

$$\mathbf{K} = \begin{pmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_{22} \end{pmatrix}, \tag{S128}$$

where

$$\begin{aligned} \mathbf{K}_{11} &= \mathbf{X}_{\mathcal{A}}^{*\top} \mathbf{R}^{-1} \widetilde{\mathbf{X}}_{\mathcal{A}}, \\ \mathbf{K}_{21} &= \mathbf{Z}_{\mathcal{B}}^\top \mathbf{R}^{-1} \widetilde{\mathbf{X}}_{\mathcal{A}} = \mathbf{K}_{12}^\top, \\ \mathbf{K}_{22} &= \mathbf{Z}_{\mathcal{B}}^\top \mathbf{R}^{-1} \mathbf{Z}_{\mathcal{B}} + \mathbf{G}^{-1}_{\mathrm{working}}. \end{aligned} \tag{S129}$$

For the jth slope, let $\mathbf{e}_{j+1}^{(P+1)}$ denote the corresponding unit vector in the fixed-effects space. Solving

$$\mathbf{K} \begin{pmatrix} \mathbf{k}_{\mathcal{A}} \\ \mathbf{k}_{\mathcal{B}} \end{pmatrix} = \begin{pmatrix} \mathbf{e}_{j+1}^{(P+1)} \\ \mathbf{0}_M \end{pmatrix} \tag{S130}$$

via the Schur complement

$$\mathbf{S} = \mathbf{K}_{11} - \mathbf{K}_{12}\mathbf{K}_{22}^{-1}\mathbf{K}_{21} \tag{S131}$$

gives

$$\mathbf{k}_{\mathcal{A}} = \mathbf{S}^{-1}\mathbf{e}_{j+1}^{(P+1)}, \qquad \mathbf{k}_{\mathcal{B}} = -\mathbf{K}_{22}^{-1}\mathbf{K}_{21}\mathbf{k}_{\mathcal{A}}. \tag{S132}$$

By the orthogonality condition above, $\mathbf{K}_{21}$ has only one potentially nonzero column, corresponding to the intercept:

$$\mathbf{K}_{21} = \big(\mathbf{c}, \mathbf{0}, \ldots, \mathbf{0}\big), \qquad \mathbf{c} := \mathbf{Z}_{\mathcal{B}}^{\top}\mathbf{R}^{-1}\mathbf{1}_N. \tag{S133}$$

Moreover, for every $j = 1, \ldots, P$,

$$(\mathbf{K}_{11})_{1,j+1} = \mathbf{1}_N^{\top}\mathbf{R}^{-1}\widetilde{\mathbf{X}}_{\mathcal{A},j} = \sigma_{\varepsilon}^{-2}\sum_{m=1}^{M}\mathbf{1}_{N_m}^{\top}\widetilde{\mathbf{X}}_{\mathcal{A},mj} = 0 = (\mathbf{K}_{11})_{j+1,1}. \tag{S134}$$

Hence, $\mathbf{K}_{11}$ is block diagonal in the intercept and slope coordinates:

$$\mathbf{K}_{11} = \begin{pmatrix} k_{00} & \mathbf{0}^{\top} \\ \mathbf{0} & \mathbf{K}_{11}^{*} \end{pmatrix}, \qquad k_{00} := \mathbf{1}_N^{\top}\mathbf{R}^{-1}\mathbf{1}_N. \tag{S135}$$

Because $\mathbf{K}_{21}$ has only one potentially nonzero column, the matrix $\mathbf{K}_{12}\mathbf{K}_{22}^{-1}\mathbf{K}_{21}$ has a potentially nonzero entry only at position $(1, 1)$, equal to

$$d := \mathbf{c}^{\top}\mathbf{K}_{22}^{-1}\mathbf{c}. \tag{S136}$$

It follows that

$$\mathbf{S} = \begin{pmatrix} k_{00} - d & \mathbf{0}^{\top} \\ \mathbf{0} & \mathbf{K}_{11}^{*} \end{pmatrix} \tag{S137}$$

is also block diagonal in the intercept and slope coordinates, and so is $\mathbf{S}^{-1}$.

Since $\mathbf{e}_{j+1}^{(P+1)}$ has a zero in the intercept coordinate for every slope index j, it follows that

$$(\mathbf{k}_{\mathcal{A}})_1 = 0. \tag{S138}$$

Consequently,

$$\mathbf{k}_{\mathcal{B}} = -\mathbf{K}_{22}^{-1}\mathbf{K}_{21}\mathbf{k}_{\mathcal{A}} = -\mathbf{K}_{22}^{-1}\mathbf{c}(\mathbf{k}_{\mathcal{A}})_1 = \mathbf{0}. \tag{S139}$$

Therefore, the random-effects component of $\mathbf{K}^{-1}\mathbf{e}_{j+1}^{(P+1+M)}$ vanishes for every fixed-effect slope β_j. Hence,

$$\mathbf{e}_{j+1}^{(P+1+M)\top}\mathbf{K}^{-1}\mathbf{A}\mathbf{K}^{-1}\mathbf{e}_{j+1}^{(P+1+M)} = 0, \tag{S140}$$

which implies

$$\kappa^2_{\text{Bayesian},j} = \kappa^2_{\text{freq},j} \qquad \text{for every } j = 1, \ldots, P. \tag{S141}$$

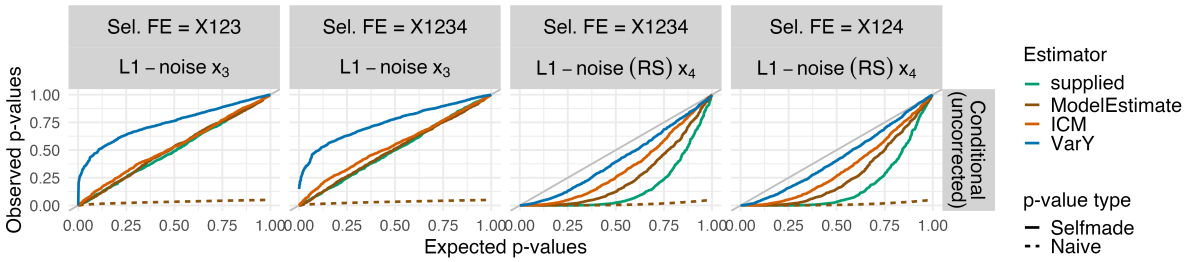
□

Simulation Study 3: Uncorrected Bayesian Conditional

Figure S5 presents Q–Q-plots of the repetition of Simulation Study 3 under the uncorrected Bayesian conditional perspective. Convergence issues were again negligible.

Figure S5

Simulation Study 3, Uncorrected Bayesian Conditional Perspective: Distribution of p-Values



Note. The figure shows Q–Q plots from a repetition of Simulation Study 3 under the Bayesian conditional perspective. Columns display observed p-values for the fixed effects of the Level-1 working noise variables (x_3 and x_4) against standard uniform quantiles. Line types distinguish naïve from selfmade p-values, and colors indicate covariance estimators. Deviations toward the lower right (upper left) indicate liberal (conservative) p-values.

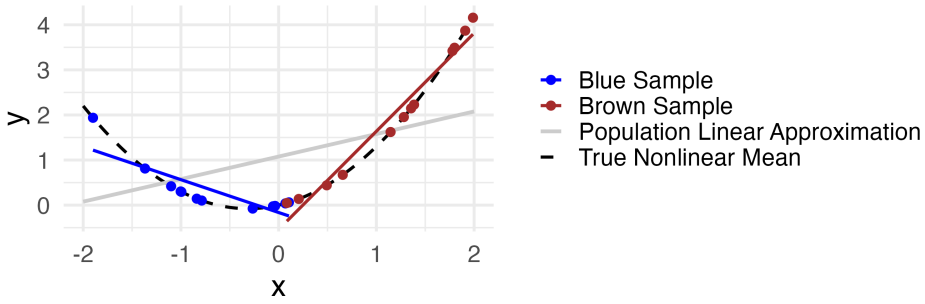
Simulation Study 3 under Random X

Motivation

As discussed in the main text, the distinction between fixed and random **X** can become important under first-order incorrectness. Different realizations of the predictor matrix may induce different linear approximations of the underlying data-generating mechanism, thereby introducing additional sampling variability (Berk et al., 2014; Buja et al., 2015). Because Simulation Study 3 used fixed **X**, we repeated selected conditions with random **X** to examine how the covariance matrix estimators perform in the presence of this additional source of variability.

Figure S6

Random Predictor Values and a Nonlinear Mean Function



Note. Adapted from Berk et al. (2014). The dashed curve represents the true DGM, and the gray line its linear population approximation. Sampling different predictor values from the same DGM (blue vs. brown) yields different fitted linear relationships.

Method

The design closely followed that of Simulation Study 3. Unlike in Simulation Study 3 however, a new Level-1 predictor matrix was generated independently for each simulated sample. Specifically, the four Level-1 predictors for each observation were drawn independently from a multivariate normal distribution with mean vector $2 \cdot \mathbf{1}_4$ and covariance matrix $\mathbf{I}_4$. Thus, the predictor values were independently generated across observations and predictors, rather than generating a single Level-1 pattern and replicating it across Level-2 units. The fixed-effects design matrix combined these predictors with their squared terms. Random effects followed the same distribution as in Simulation Study 3. We specified the Level-1 residual variance in two ways. In scenario (a), we analytically derived the population

variance of the fixed-effects predictor for a randomly selected observation n, denoted by $\hat{y}_n$, as

$$
\begin{aligned}
\operatorname{var}(\hat{y}_n) =& (-0.5)^2 \operatorname{var}(x_{1n}^2) + 0.5^2 \operatorname{var}(x_{2n}^2) \\
&+ (-4)^2 \operatorname{var}(x_{3n}) + \operatorname{var}(x_{3n}^2) \\
&+ 4^2 \operatorname{var}(x_{4n}) + \operatorname{var}(x_{4n}^2) \\
&+ 2 \cdot (-4) \cdot 1 \cdot \operatorname{cov}(x_{3n}, x_{3n}^2) + 2 \cdot 4 \cdot (-1) \cdot \operatorname{cov}(x_{4n}, x_{4n}^2) \\
=& 0.25 \left(2 \operatorname{var}(x_{1n})^2 + 4\mathbb{E}(x_{1n})^2 \operatorname{var}(x_{1n})\right) \\
&+ 0.25 \left(2 \operatorname{var}(x_{2n})^2 + 4\mathbb{E}(x_{2n})^2 \operatorname{var}(x_{2n})\right) \\
&+ 16 \operatorname{var}(x_{3n}) + \left(2 \operatorname{var}(x_{3n})^2 + 4\mathbb{E}(x_{3n})^2 \operatorname{var}(x_{3n})\right) \\
&+ 16 \operatorname{var}(x_{4n}) + \left(2 \operatorname{var}(x_{4n})^2 + 4\mathbb{E}(x_{4n})^2 \operatorname{var}(x_{4n})\right) \\
&- 8 \left(2\mathbb{E}(x_{3n}) \operatorname{var}(x_{3n})\right) - 8 \left(2\mathbb{E}(x_{4n}) \operatorname{var}(x_{4n})\right) \\
=& 13.
\end{aligned}
\tag{S142}
$$

Combining this result with the same pseudo coefficient of determination, $R^2_{\text{pseudo}} = 0.3$, yielded $\sigma_\varepsilon^2 = 10.53$ according to (37). In scenario (b), we set the Level-1 residual variance to the smaller value $\sigma_\varepsilon^2 = 0.25 \cdot \operatorname{var}(\hat{y}_n) = 3.25$. Finally, we generated the outcomes according to the LMM specified in Simulation Study 3.

We evaluated the selfmade approach when the full linear model was selected under both the efficient and not-efficient marginal perspectives, using either the true model-implied covariance matrix $\boldsymbol{\Sigma}$ or the *ModelEstimate* estimator, with standard or CR2 variance estimation. Under the not-efficient marginal perspective, we additionally evaluated the newly proposed robust covariance matrix estimators. The results are based on 10,000 Monte Carlo replications.

For the not-efficient marginal perspective, the population working-model coefficients of x_3 and x_4 can be verified analytically. Because the predictors are mutually independent, the

OLS projection coefficient for x_3 is

$$\begin{aligned}\frac{\operatorname{cov}(x_3, -4x_3 + x_3^2)}{\operatorname{var}(x_3)} &= -4 + \frac{\operatorname{cov}(x_3, x_3^2)}{\operatorname{var}(x_3)} \\ &= -4 + \frac{2\mathbb{E}(x_3)\operatorname{var}(x_3)}{\operatorname{var}(x_3)} = -4 + 2\mathbb{E}(x_3) = 0,\end{aligned} \tag{S143}$$

where $\mathbb{E}(x_3) = 2$. Similarly,

$$\begin{aligned}\frac{\operatorname{cov}(x_4, 4x_4 - x_4^2)}{\operatorname{var}(x_4)} &= 4 - \frac{\operatorname{cov}(x_4, x_4^2)}{\operatorname{var}(x_4)} \\ &= 4 - \frac{2\mathbb{E}(x_4)\operatorname{var}(x_4)}{\operatorname{var}(x_4)} = 4 - 2\mathbb{E}(x_4) = 0.\end{aligned} \tag{S144}$$

Thus, x_3 and x_4 are population working-noise predictors under the not-efficient marginal perspective. To assess whether the corresponding GLS working-model coefficients are also effectively zero under the efficient marginal perspective, we generated 10,000 samples without model selection and computed the mean observed statistics t_{obs} for both predictors. The means were close to zero across covariance specifications (Table S1), with Monte Carlo standard errors ranging from 0.00123 to 0.00256. These results are consistent with GLS working-model coefficients close to zero, although they do not establish that the population targets are exactly zero. Thus, testing $H_0 : \rho = 0$ for x_3 and x_4 is supported under both marginal perspectives.

Table S1

Mean observed statistics t_{obs} and Monte-Carlo standard error for x_3 and x_4 based on 10,000 simulations without model selection.

Marginal perspective	Covariance specification	$\bar{t}_{\text{obs},x_3}$	MCSE_{x_3}	$\bar{t}_{\text{obs},x_4}$	MCSE_{x_4}
Not-efficient	True $\boldsymbol{\Sigma}$/*ModelEstimate*	0.00064	0.00162	0.00285	0.00256
Efficient	True $\boldsymbol{\Sigma}$	−0.00083	0.00123	0.00775	0.00233
Efficient	*ModelEstimate*	−0.00120	0.00129	−0.00503	0.00239

Note. MCSE denotes the Monte Carlo standard error of the mean observed statistic, computed as the empirical standard deviation divided by $\sqrt{10{,}000}$.

Results

Figure S7 shows Q–Q plots of the post-selection p-value distributions in scenario (a) and (b). Under both marginal perspectives, supplying the true model-implied covariance matrix resulted in liberal p-values. The liberality was more severe in scenario (b). Moreover,

the deviation from uniformity was more pronounced for the fixed effect associated with a random slope in the DGM (β_4). Using *ModelEstimate* instead produced *p*-values that were closer to the uniform distribution, although non-negligible deviations remained. Under the efficient marginal perspective, using CR2 variance estimation with the *ModelEstimate* covariance matrix estimator resulted in approximately uniformly distributed *p*-values. The same was observed under the not-efficient marginal perspective when our proposed covariance matrix estimators were used. Under the not-efficient marginal perspective (fourth row in Figure S7), our implementation of CR2-based variance estimation resulted in slightly liberal *p*-values for the fixed effect of x_4 in both scenarios.

Discussion

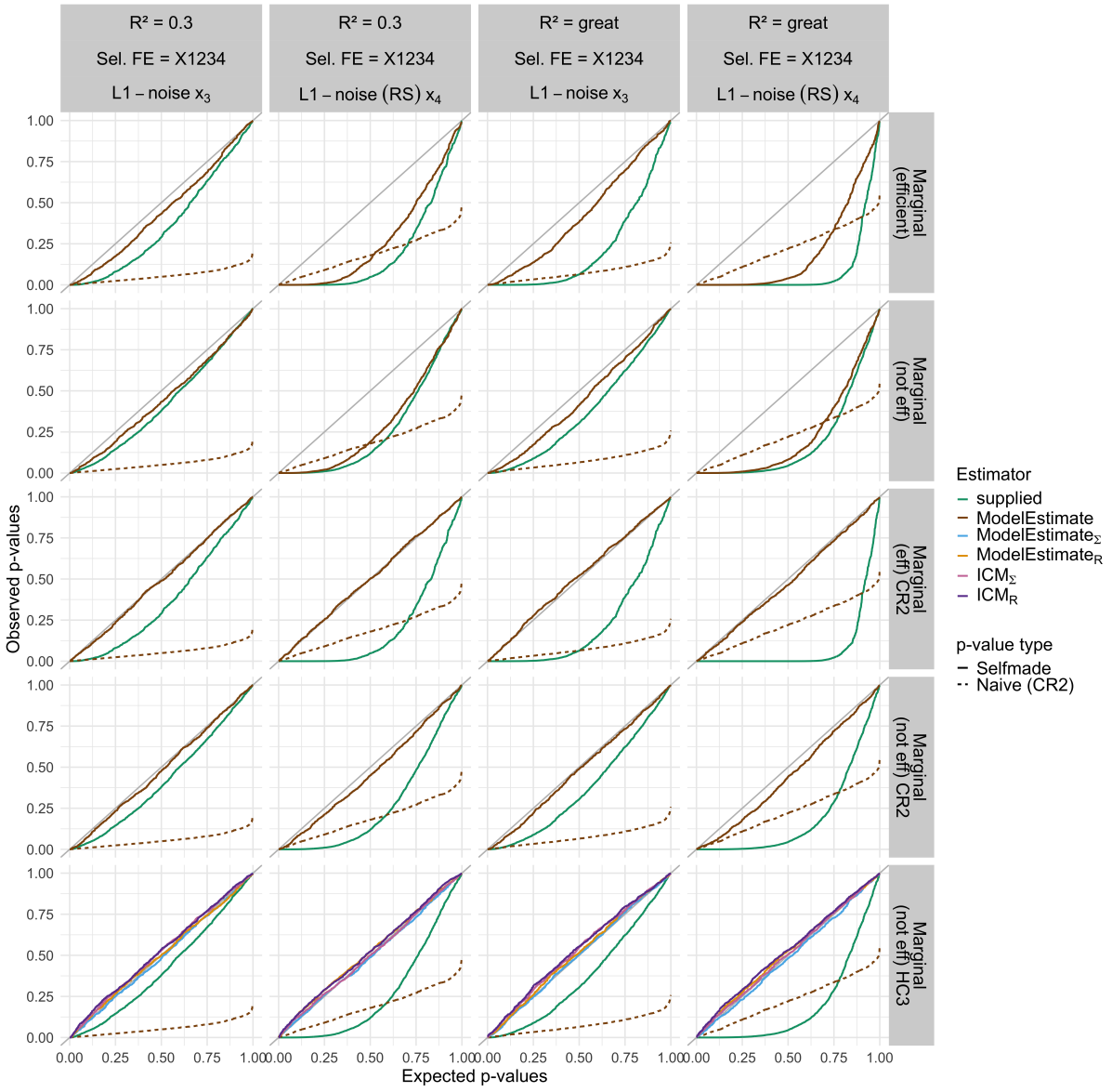
Unlike in Simulation Study 3, where **X** was fixed, supplying the true model-implied covariance matrix did not yield valid post-selection inference when **X** was random. Under the misspecified selected model, the sample-specific linear projection of the mean function varies with the realization of **X**. The supplied covariance matrix accounts for the sampling variability of **y** conditional on the realized **X**, but does not account for the additional variability of this projection across realizations of **X** when testing a population-level parameter (e.g., Buja et al., 2015). The *ModelEstimate* covariance matrix estimator proposed by Rügamer et al. (2022) likewise failed to provide valid post-selection inference, reducing the practical usefulness of the original selfmade approach.

By contrast, our proposed covariance matrix estimators, which are based on HC3 standard errors, as well as the CR2-based approaches, improved the calibration of the post-selection *p*-values under the misspecified selected model. This is consistent with the robustness of sandwich standard errors to first-order incorrectness under random **X** (Buja et al., 2015; White, 1980). Combined with these robust approaches, the selfmade approach yielded approximately valid post-selection inference under the examined random-**X** conditions, despite the misspecification of the selected working model. In particular, under the efficient marginal perspective, CR2-based variance estimation in combination with the *ModelEstimate* covariance matrix estimator yielded approximately uniformly distributed *p*-values. Reducing the Level-1 residual variance in scenario (b) produced more liberal selfmade *p*-values under

standard covariance matrix estimators because the consequences of first-order incorrectness became more pronounced relative to the residual noise. Nevertheless, both the proposed HC3-based covariance matrix estimators and the CR2-based approaches continued to provide approximately valid post-selection inference under this more challenging condition. We therefore conclude that combining the selfmade approach with robust approaches to non-sphericity of $\mathbf{R}^*$ can extend approximately valid post-selection inference to random-$\mathbf{X}$ settings with a misspecified selected working model, under the conditions examined here.

Figure S7

Simulation Study 3 under Random X: Distribution of p-Values



Note. The figure shows results from a repetition of Simulation Study 3 with a random predictor matrix. $R^2 = 0.3$ corresponds to scenario (a), and $R^2 =$ great to scenario (b). Columns display observed p-values for the Level-1 working noise variables (x_3 and x_4) against standard uniform quantiles. Rows distinguish the inferential perspectives used in the selfmade approach. For readability, additional rows show the p-value distribution under the robust approaches. Colors distinguish the covariance matrix estimators. Deviations toward the lower right (upper left) indicate liberal (conservative) p-values.